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From the observations of Tycho Brahe to GW250114: three results in physics re-measured with the answer withheld

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We ask whether AI agents can re-measure historical results of physics from the primary records under a written protocol that keeps the expected answer out of the agents' instructions and fixes in advance the comparison with modern values. In practice each analysis also fixed its plan before the reported result was computed, froze the result before that comparison (by hash in episode 1 and, for the decision-setting result, in episode 5; in episode 3 only by commit order, beside a control carrying the same physics, and its headline was refitted after the checker's comparison), passed each measurement to a separate re-analysis agent and kept a record of every run; Section 2 says where the record falls short. The paper contains three measurements, one comparison and one feasibility test. Given Tycho Brahe's Mars logs and no model family, an agent chose two Keplerian orbits about one common point, which predicted the withheld years 1596–1601 to a median of $2.46'$. Newcomb's Mercury transit contacts of 1677–1881, fitted with a Newtonian model, require an extra perihelion advance of $43.71 \pm 1.04''$ per century, 0.7σ above general relativity. For GW250114 every one of the 40 analysis cells, over two independently run inspiral-only posteriors, gave $P(\Delta A > 0)$ of at least 0.846; once a single-mode bias in the earliest ringdown fits had been found and those cells withdrawn, the retained cells of the headline lane gave $P = 0.86$–0.98, with a minimum of 0.86 at 20 M_f, below the pre-registered 0.90, so the rule withheld the claim. A brief that leaked the shape of the answer, an offset attributed to the wrong cause, the single-mode bias and a failed run were each caught and reported.

1. INTRODUCTION

The results that shaped celestial mechanics and gravitation were measured from records that still exist. Tycho Brahe's Mars observations survive in his logs, printed in Dreyer's edition [1]. Le Verrier (1859) [2] and Newcomb (1882) [3] found an unexplained perihelion motion of Mercury, and the transit contact times of 1677–1881 are tabulated in Newcomb's 1882 discussion. The strain data of the black-hole merger GW250114 are public. Each of these measurements can be redone from the record. Redoing them by hand is slow. Redoing them without bias is harder, because the analyst knows the answer.

We ask whether AI agents can redo such measurements to a standard a physicist would accept, and what that standard must be when the analyst knows the answer. The claim is methodological and rests on three measurements, each reviewed by a checker. In each, agents carried a measurement from the primary record to a result that a separate re-analysis agent reproduced; for GW250114 the checker's report predates the final result, which a further independent review recomputed from the committed samples (Section 2). The written protocol required that the agents' instructions withhold the expected answer and its shape, and fixed in advance the comparison with modern values, naming $42.98''$ for Mercury and the GW150914 targets; the agents who read the protocol therefore saw those numbers. In practice the analyses also fixed their plans before the reported results were computed, froze each result before that comparison and kept a record of every run, with the exceptions stated in Section 2. The third, of the horizon area of GW250114, ends in a pre-registered decision not to claim that the area grew. The agents were not ignorant of the answers, and we do not claim they were. Our claim is that the protocol limits what that knowledge can do and that its record shows where it failed.

We use four terms in a narrow sense. *Answer-withheld* means that the instructions named neither the expected value nor its form. Runs that also withheld the data the comparison depends on, namely the held-out years of the Kepler fit and the disguised records, we call *held-out* runs. *Checker* is the separate AI agent that re-ran or, for GW250114, re-derived each result and compared it with modern values; its reports are filed as referee reports, in the repository for episodes 1 and 3 and with the manuscript for episode 5. The protocol calls the author's role referee, and we call it *arbiter*. The arbiter reviews results and method and does not steer toward the known answer. The author set the project's goal and approved the protocol. An orchestrating agent drafted the protocol and the briefs and ran the analysis lanes, and other agents transcribed, reduced, fitted and checked.

The evidence consists of three measurements. From Tycho's logs, agents produced a 187-position data product and six fits of it: one under prescribed models, one with no model family named and the later years withheld, and four on disguised copies. From Newcomb's 58 concluded transit contacts, an agent measured Mercury's excess perihelion advance and compared it with general relativity, a comparison the protocol had fixed in advance. For GW250114 agents combined two independently run

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inspiral-only posteriors with their own ringdown fits under a pre-registered decision rule and calibrated the method on GW150914. The Supplementary Information reports a feasibility test of Newton's Moon test and Jupiter-satellite exponent. It has no plan or checker and is not counted.

The failures are part of the evidence. A brief leaked the shape of the answer, and the checker graded the leak a blocker. When a fitter attributed an offset to Tycho's data, the checker showed that the data were not at fault. A run failed and was reported with its declared rerun. The strongest GW250114 cells rested on a known single-mode bias; the checker flagged it, the pre-registered overtone test passed, and those cells were withdrawn. Section 4 lists every such event, who caught it and what changed.

Machine reanalyses of historical science exist (Supplementary Section S4), and most start from Kepler's already reduced tables or from synthetic data. The closest work is KeplerBench (Malhotra 2026) [4], a benchmark for discovering physical laws from observer-centred astronomical measurements. Its pilot on Tycho's Mars record already reports the qualitative finding of our model-agnostic Kepler fit, that a flexible fit matches the training years and fails on the later ones while a Keplerian orbit does not. The pilot uses Tycho's Mars declinations as transcribed by Pafko [5], 354 nightly epochs from 1582 to 1600 with the 268 before 1594 for training and 86 held out. Its ellipse, fitted with the modern Earth orbit supplied, has a held-out median residual of 13.31′. Our Kepler measurement differs in the level of the data, raw instrument readings reduced to longitude and latitude with per-session errors, and in fitting Earth's orbit from the same record. It also differs in letting an agent choose its own model families under a pre-registered plan with the later years withheld, and in the disguised runs and the checker's re-analyses. Our held-out median of 2.46′ is an angular distance on different data and a different split, so the two medians are not directly comparable. We could not find a reduction of the raw readings to longitude and latitude with per-session errors, a joint fit of Earth's and Mars's orbits from the same record, a refit of Newcomb's 1882 contact compilation with a numerically integrated model, or agents run on raw historical measurement records. For GW250114 our analysis is a transparent replication, not a new test.

Section 2 states the protocol, Section 3 the results, Section 4 what the protocol caught, Section 5 the limitations and Section 6 the conclusions; Appendix A gives the methods, Appendix B the Extended Data figures and Appendix C the Supplementary Information.

2. THE PROTOCOL

The protocol was written on 13 October 2026 and committed at 20:59 local time (UTC−6, as are all local times in this paper), after the kill tests of episodes 1 and 3 had begun (20:33) and before any batch transcription, batch reduction or fit. Those kill tests reduced two nights of Tycho's observations and compared Newcomb's 58 contacts with Horizons; neither fitted anything about the planets' motion. Episode 5's first step, an area change computed from the public full-signal posteriors, also ran before the commit (20:35–20:58), and no result in this paper uses it; its first ringdown fit began at 21:08. The protocol gives agents the data and a question but never the expected answer or its shape. The briefs say "find the orbit that fits these positions" rather than "fit an ellipse", and ask "what drift of Mercury's perihelion is left after the planets?" rather than "find 43″". A run that breaks this rule is labelled exploration and repeated. The one run that broke it was repeated and is reported as a measurement under prescribed models, not as exploration (Section 3.1). Every number must trace to a checksummed data file, a code commit and a run log. Data are never committed; each episode carries a fetch script and a checksum list. Inputs taken from the literature are marked "cited". The comparison of each result with modern values was fixed in advance: Kepler's orbit against the JPL Horizons ephemeris (Mars barycentre, DE441), Mercury's excess advance against general relativity's 42.98″ per century, and the area result against Isi et al. (2021) [6] on GW150914 and the LIGO-Virgo-KAGRA (LVK) analysis on GW250114 [7]. Keeping a record of every run is the project's practice rather than a written rule, and the paper does not describe every run: the trial fits of the model-agnostic Kepler fit, two re-run attempts by its checker that were stopped after about ten minutes, and most of the GW150914 conditioning fits are recorded only in the repository; the log of the Mercury fit that crashed (`data/fit_log_attempt1.txt`) survives only in the working copy, because `data/` is git-ignored.

Each analysis began with a kill test with a pass criterion: for Tycho, agreement with Horizons within 5′ on two nights transcribed by hand and reduced by script, plus a legibility count; for Mercury, a contact scatter below 10″ along the planet's path. The analysing agent then wrote a plan fixing models, outlier rule, uncertainty recipe and decision rule. The evidence that the plan came first varies in strength. In episode 3 the plan's file time is 21:17 and the first historical fit log's 21:35, both working-copy times; plan and result entered the repository in one commit at 22:13. In episode 1 the plan, code and result entered the repository in one commit, so the order rests on the plan's own statement and the run logs; in the model-agnostic fit, short trial fits preceded the first full run, and the plan's amendments cannot be dated against them. From status section S6 on, each pre-registration was saved as a numbered version of the status file, with its sha256 and a copy, before the quantity it governed was computed. S5, the first entry marked pre-registered, carries only its own timestamp, and the repository dates it after its fits. Every 220-only fit in the decision set predates S5: the baseline fits at 8–20 M_f ran on the first evening, and the no-high-pass and PE-spectrum variants in a session that same evening whose subject was the gap between our 220 fits and LVK's published ones; by then their remnant masses had been compared with the full-signal remnant

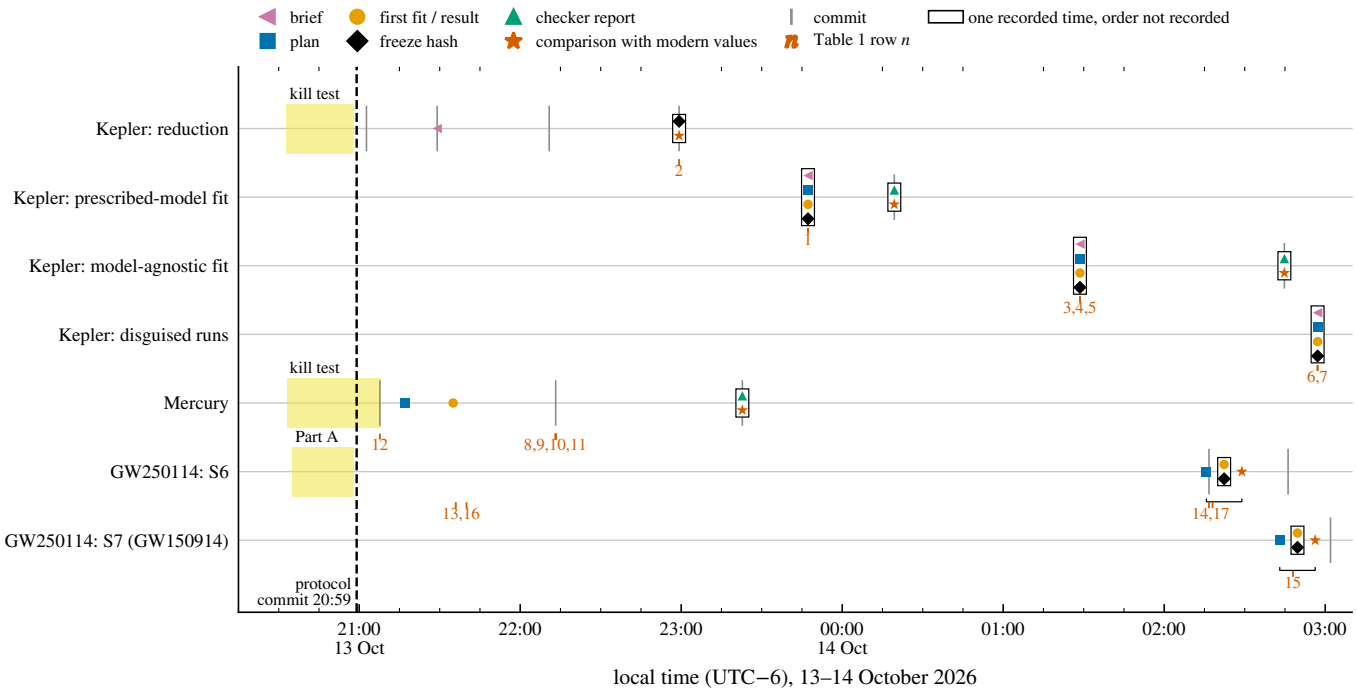


FIG. 1. **The protocol as executed.** One row per measurement (Kepler: reduction, prescribed-model fit, model-agnostic fit, disguised runs; Mercury; GW250114: S6 and S7) on a common time axis for 13–14 October 2026. Markers: brief, plan, first fit, freeze hash, checker report, comparison with modern values. The protocol commit (20:59) is a vertical line; the three steps that began before it (the kill tests of episodes 1 and 3 and episode 5's Part A) are shaded. Markers stacked in a box share one recorded time (one commit, or one entry of the status file), and the record does not fix their order; markers joined by a bracket carry their own recorded times, minutes apart, and are spread in that order. No brief time is recorded for Mercury or for GW250114. Each row of Table 1 is marked where it occurred; rows 18–22 were recorded on 14 October after the window shown, rows 18–21 at 09:02–09:45 local time (episode 5, status sections S9 and S10) and row 22 at 10:48–10:49 (status section S11).

and with LVK's 220 fits. When the S6 decision set was fixed, the ringdown side of every cell was known and the inspiral-only side was not. $P_{\min}$ and the 0.98 upper end are no-high-pass cells; the baseline cells alone give $P_{\min} = 0.864$ and the same sentence, and the 220+221 cells alone, the only ones pre-registered before fitting, give 0.892 at 13 M_f, the same sentence, though with $P_{\min}$ + 2 s.e. at the 0.90 boundary. Of the GW250114 ringdown fits, only the overtone-test fits (220+221, with 220+330 beside them) were pre-registered before fitting.

Freezing varied as well. Each episode-1 fitter wrote the sha256 of its result file to FROZEN.txt, after which the result file did not change. Episode 5 recorded the sha256 of each status version before the next step; the pre-registration copies are committed, the hashed result versions are not, so the result freeze rests on the recorded hash alone. The overtone cells, the second lane and the later checks (S9–S11) were pre-registered after the comparison with LVK and could not change the sentence, which $P_{\min}$ at 20 M_f sets. Episode 3 has a timestamped plan but no frozen hash. Additions made after a result was seen are labelled post hoc.

After the freeze, the checker received read-only access to the repository, re-ran the code and compared the result with modern values, grading its findings as blocker, should-fix or polish. The checker has done this for the two refereed Kepler fits, for Mercury and for GW250114; the disguised Kepler runs and the Newton feasibility test have no checker report yet. In episode 1 the fixes the checker required changed wording and disclosures but no numbers. In episode 3 they re-derived the headline from a converged fit (43.706 ± 1.036; the stalled run had given 43.7 to 43.8 ± 1.0). In episode 5 its verdict was that the headline stands, with changes; they led to an overtone test, an autocorrelation-aware standard error, a fragility check and the withdrawal of the strongest cells, and the headline sentence did not change. The checker's report covers status sections S1–S8 and the headline before the overtone test; S9 and S10 were run in answer to it. A further independent review, by another re-analysis agent, then recomputed from the committed samples P for the 20 M_f and 220+221 cells, the lane A_i statistics and the f_c proxy, plus the fragility and standard-error figures. The ringdown fits were not re-run.

The protocol lists one deviation, the brief of the first Kepler fit, which named the model families and quoted Kepler's $8'$. Two more belong on the list. The disguised fits were added after the protocol was written. The GW150914 calibration of the area method was computed after the GW250114 result, although each was pre-registered before its own computation. The protocol asks that the execution logs of runs on one of the analysis platforms

be exported into a provenance folder in each episode. No such folder exists yet; the agent runs are logged to the harness and to git.

Figure 1 shows the protocol as it was executed, episode by episode, with each deviation and catch marked where it occurred.

3. RESULTS

3.1. Kepler's orbit from Tycho's logs

Six agent workers transcribed 427 Mars sessions from page images of Tycho's logs, as printed in Dreyer's edition, with a confidence on every number. The orchestrating agent reduced each session on its own to geocentric ecliptic longitude and latitude with formal errors; nothing about the motion of Mars entered the reduction. Of the 427 sessions, 349 were reduced, and the 187 sessions whose internal quality flags were good (classes A and B) form the frozen data product. Against Horizons, used only for verification, their median total residual is 1.38′, and without five gross outliers the RMS is 1.50′ in longitude and 1.78′ in latitude, consistent with Tycho's known instrumental accuracy (Wesley 1978 [8]; Extended Data Fig. 1); for comparison, Tycho's own printed positions have a median longitude residual of 1.89′ against Horizons (Methods M1). That verification is how the five gross outliers were found. They stayed in the product, because re-reading them would have leaked the ephemeris into the data.

The first fit broke the protocol. Its brief asked for "the orbit that fits" but prescribed three model families (a circle with an equant, a Mars ellipse with a circular Earth, and two Kepler ellipses) and quoted Kepler's 8′. The run is therefore a measurement under prescribed models with modern values withheld, not a discovery. Two Kepler ellipses won (ΔBIC = 9182). For Mars they give $a = 1.523697 \pm 0.000065$ Earth units, $e = 0.093065 \pm 0.000072$ and $P = 686.9736 \pm 0.0020$ d; Earth's orbit was fitted from the same data, and Kepler's third law was not imposed. The checker reproduced the fit bit for bit. The Mars elements agree within 2σ with a two-body fit to DE441 at the same epochs, except the inclination, which is 13.7″ (3.0σ) high; the checker found this consistent with a latitude offset in the reduction, a likely but not demonstrated cause. A circle with a free equant, given the same Keplerian Earth, leaves a largest median longitude residual of 17.0′ over 30° bins of Mars's mean anomaly, against 0.88′ for the ellipses, and of 7.53′ against 1.55′ over apparitions (Extended Data Fig. 2). We did not repeat Kepler's own construction and do not claim to have recovered his 8′.

The run that answers the breach told the agent nothing about model families and withheld every row after mid-January 1596, leaving 159 rows for fitting and 28 for prediction (Fig. 2). The brief required at least three structurally different families, one with no physics, and fixed

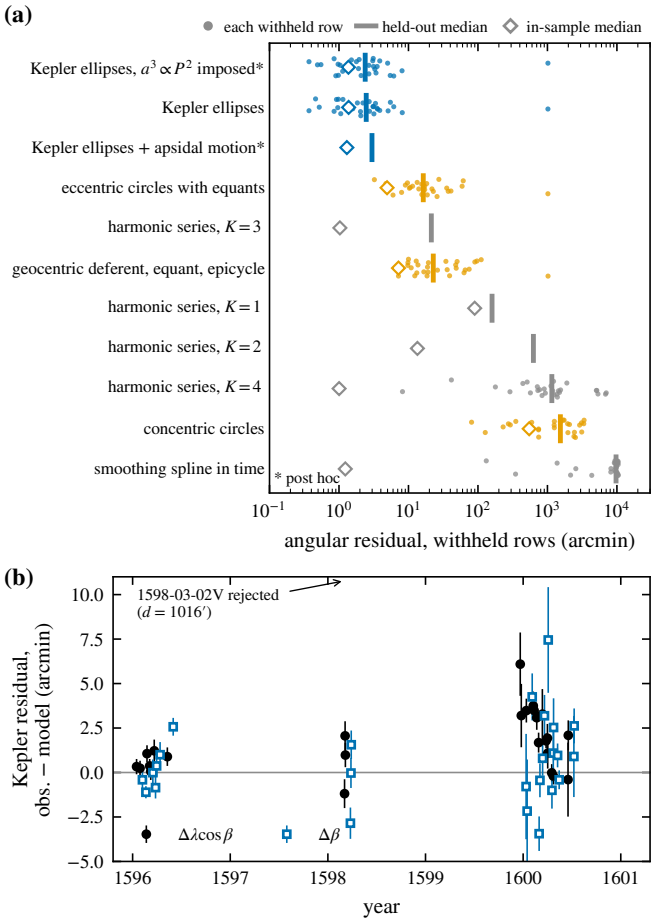


FIG. 2. **Held-out residuals by model family.** Angular residual of each of the 28 withheld rows (1596–1601) for the seven families whose per-row residuals were stored, on a log scale; for harmonic K = 1–3 and the post-hoc apsidal model only the medians exist. Lower panel, the winning model's residuals by component.

the hold-out; it left the decision rule to the agent's preregistered plan, which chose the lowest held-out median residual, with a 1.25× tie broken by BIC. Two Keplerian ellipses about one common point predicted the withheld years to a median of 2.46′. Eccentric circles with equants reached 16.3′ (15.4′ from the checker's wider multistart), and the best no-physics model, a harmonic series, fitted the training years to about 1′ but missed the withheld years by 21′. With the size ratio free, a^3/P^2 is the same for both orbits to 0.99991 ± 0.00012, equivalent to an exponent $q = 0.66662 \pm 0.00006$ in $a \propto P^q$, consistent with Kepler's 2/3 at -0.7σ. With two orbits this measures the exponent; it does not test a power law. Directions alone cannot tell a Sun-centred system from a Tycho-style one with the same relative motion, so the result is two Keplerian motions about one common point. KeplerBench's pilot (Malhotra 2026) [4] reports the same qualitative pattern on Tycho's Mars declinations, where a flexible curve fails on the later campaigns and an ellipse supplied with the modern Earth orbit degrades from 2.66′ in sample to 13.31′ held out. In this run the agent chose the families

itself under a pre-registered plan, and the positions were reduced from the raw readings with Earth's orbit fitted from the same record. Our residuals are angular distances combining longitude and latitude; theirs are in declination alone, on a different split; 2.46′ and 13.31′ are therefore not directly comparable.

The author asked whether the agent was cheating, that is, whether it finds ellipses because it recognises Mars. To test this, fresh agents received copies of the product with time rescaled, longitude rotated and optionally mirrored, latitude optionally flipped and every label removed. Three disguised copies took four runs. Three runs, one on each copy, found two Kepler ellipses, with held-out medians over the unflagged rows of 2.05′, 2.13′ and 2.02′ (the undisguised fit gives 2.21′ on the same basis, 2.46′ over all 28 rows). Each identified the data as Mars seen from Earth only after the fit, from the fitted invariants. The fourth run, on an overloaded machine, failed to reach the two-ellipse basin; the declared rerun on the same copy is the run that gave 2.13′ (Supplementary Section S2). The disguise shows that recognition followed the fit. It cannot show that knowledge played no part in which families the agents chose to try.

3.2. Mercury's excess perihelion advance, and the comparison with general relativity

The record is Newcomb's 58 concluded geocentric contacts from 23 transits of 1677–1881, with the weights he printed. The instructions named no expected value and no theory. No brief file survives for this episode; the statement is the measurement report's own. The model is a Newtonian N-body integration with modern masses, with the Earth's and the Sun's positions taken from Horizons, and it carries an extra uniform perihelion advance D as an inverse-cube central acceleration. The contacts require $D = 43.71 \pm 1.04''$ per century with the pre-registered nuisance model, one time offset per contact type (Fig. 3). That nuisance model is nearly degenerate with Mercury's mean longitude, because transits sample the orbit at two longitudes only. Its first fit crashed and its main run stalled. A better-conditioned model with one offset of the contact separation per type, added post hoc and labelled as such, gives 43.53 ± 0.84. Every variant tried lies between 43.2 and 44.3. Applied to Horizons' own contact instants, the same pipeline returns 43.007, the relativistic advance plus the Sun's oblateness that the ephemeris contains. This control validates the pipeline to about $0.01''$ per century but says nothing about the historical data.

The checker reproduced the committed results file exactly from the committed code. The headline changed after the checker's report. The main run had stalled between 43.67 and 43.78 (43.78 ± 1.02 at its linear-step point); a wide re-fit was stopped after 11 iterations at 43.69–43.72, and 43.7 ± 1.0 was committed from a hand reading of that log. The checker found that the headline traced only to the log of that stopped run, and the converged refit committed with its report gives 43.706 ± 1.036: unchanged from the committed 43.7 at that precision, $0.07''$ below the linear-step point, with σ 1.02 → 1.04. It also found that the historical fits had started from the control's D of 43.007; refits from $D = 0$ reach the same values (43.532 ± 0.837 radial; 43.63–43.69 $\pm$ 1.02 time form, the last evaluated point and the final linear step). The "error in Newcomb's 1677 reduction" reported by the kill test was traced by the checker to Newcomb's own printed corrigendum, which our reduction had recovered independently.

The comparison with general relativity was fixed in advance against $42.98''$ per century. The first comparison that names general relativity is in the checker's report; the control (43.007 with time offsets, 43.008 with radial offsets), whose physics includes the relativistic advance and the Sun's oblateness, was fitted first, seeded the historical fits, and is compared with them in the same commit as the first fit. Fitted to Newcomb's 58 contacts, the excess advance is $43.71 \pm 1.04''$ per century, 0.7σ above the $42.98''$ general relativity predicts (Will 2014 [9]; cited). The historical contacts fix the excess advance to 2.4% (1σ); within that precision they agree with the prediction and with the same pipeline run on the DE441 ephemeris ($43.01''$ per century, relativity plus the Sun's oblateness). Newcomb's 1882 value from the same contacts, $42.95''$, rests on a Venus mass that raises his planetary Venus term by about $2.4''$ per century, so it is not on the same footing. Morrison and Ward (1975) [10], fitting about 2400 contacts of 1677–1973 with the analytical theories of their day, found $+41.9 \pm 0.5''$ per century, $1.8''$ (1.6σ combined) below our value. The two determinations differ in data span, orbital theory and ΔT treatment, and ours rests on Newcomb's concluded times.

3.3. The horizon area of GW250114

Hawking's area theorem [11] requires the remnant area A_f to be at least the sum A_i of the initial areas; LVK report $A_f > A_i$ at 4.4σ [7] on a Gaussian statistic. We asked whether this holds with our own ringdown fits and the exact one-sided tail probability $P(\Delta A > 0)$ in place of a Gaussian approximation. A_i comes from two inspiral-only posteriors (20–131 Hz) produced by two independent runs of the same analysis (lanes 1 and 2), and A_f from our fits of the 220 ringdown mode at start times of 8 to 20 M_f after the peak and, in a pre-registered test, of the 220 mode with the 221 overtone at 10.5 to 18 M_f. Before any area was computed, a decision set was fixed, with the minimum P over it ($P_{\min}$) as the headline and four candidate sentences; every later addition was pre-registered the same way. Throughout, σ for our P is one-sided, $\Phi^{-1}(P)$.

The earliest reading did not survive. Our 220-only fits gave P of 0.998–0.999 at 10.5–13 M_f, but their remnant mass fell by 9.3 $M_\odot$ between 13 and 15 M_f, as LVK's

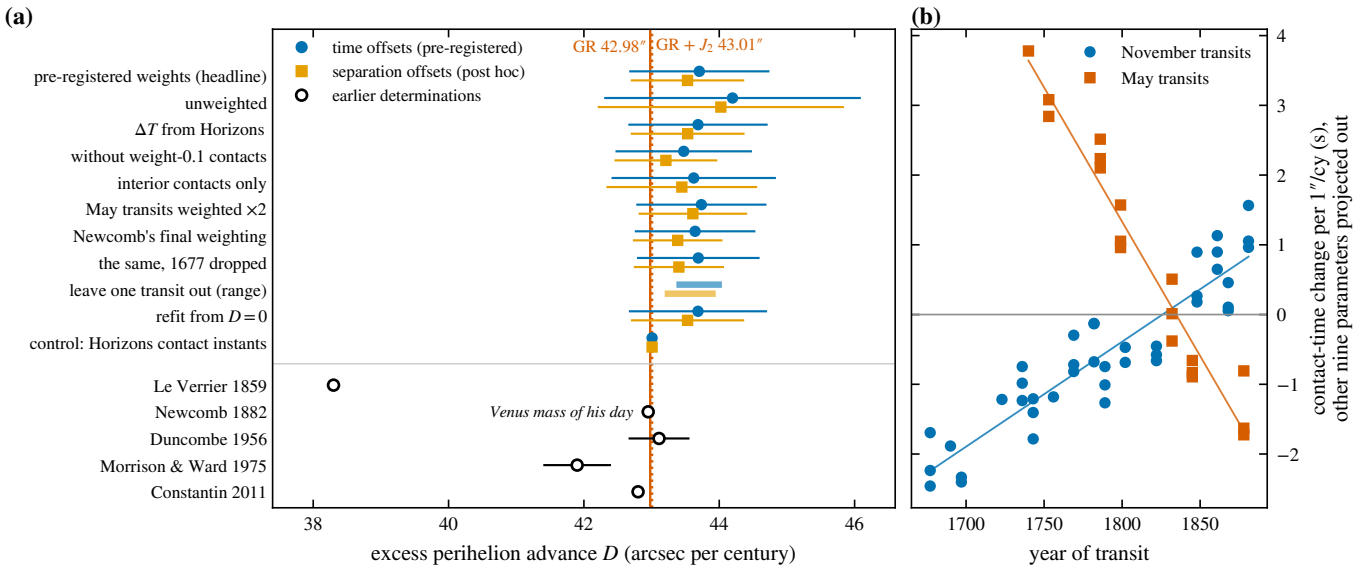


FIG. 3. **Mercury's excess perihelion advance.** D with 1σ bars for the pre-registered headline (43.71 ± 1.04), the post-hoc radial form (43.53 ± 0.84), each robustness variant, the from-zero refits and the control (43.007), with general relativity's $42.98''$ and the J_2-corrected $43.01''$ as lines, and earlier determinations as open symbols: Le Verrier 1859 (38.3), Newcomb 1882 (42.95, marked "Venus mass of his day"), Duncombe 1956 (43.11 ± 0.45), Morrison and Ward 1975 (41.9 ± 0.5) and Constantin 2011 (42.8). A second panel shows D's signature: contact-time change per $''$/cy, with opposite slopes in November and May transits. Bars are the formal, χ^2-scaled 1σ; Le Verrier, Newcomb and Constantin are drawn without bars because no uncertainty in this form is published.

own 220 posteriors do (theirs from 70.2 to 64.3 $M_\odot$). A single damped sinusoid that starts near the merger overestimates the final mass (Kastha et al. 2022 [12]), so the checker flagged these cells and we pre-registered an overtone test. With the 221 overtone, M_f falls by 12.0, 13.9 and 12.7 $M_\odot$ at 10.5, 12 and 13 M_f, agrees with the 220-only fits within 0.3 $M_\odot$ at 15 and 18 M_f, and the drop disappears (Fig. 4). The early 220-only probabilities are therefore withdrawn as a single-mode artefact. They were also fragile: 37–87% of their failing sample pairs ($A_f \leq A_i$) come from 5 extreme pre-merger samples, and removing those raises the one-sided σ by 0.14–0.55. In the cells that set the headline the top five supply at most 5.6% of the failing pairs.

The final result is sentence 4. The two lanes give A_i medians of 2.992 and 3.024 $\times 10^{11}\,\mathrm{m}^2$, a difference far below the 1σ width of 0.308 $\times 10^{11}\,\mathrm{m}^2$, so lane 1 carries the headline by the pre-registered rule. $P_{\min}$ is 0.863 for lane 1 and 0.847 for lane 2 (0.8466), both at 20 M_f, and all 40 cells across both lanes favour growth. The 220-only fits predate the pre-registration (Section 2); without the no-high-pass variant $P_{\min}$ is 0.864. The fourth pre-registered sentence, with the measured ranges inserted as the S6 and S9 pre-registrations provided, reads: whether the total horizon area of GW250114 grew depends on analysis choices; combining an inspiral-only (20–131 Hz) posterior with our own ringdown fits, $P(\Delta A > 0)$ is 0.89–0.96 for fits that include the 221 overtone and start 10.5–13 M_f after the peak, 0.91–0.98 at 15–18 M_f, and 0.86 at 20 M_f, so with our ringdown fits we cannot claim it. Allowing for autocorrelation in the ringdown chains, $P = 0.86 \pm 0.01$ at 20 M_f (the checker's larger estimate), and $P_{\min} + 2$ s.e. $= 0.885$ stays below 0.90. A proxy for a $\pm 10\%$ change of the 131 Hz cutoff, not a rerun, moves $P_{\min}$ to 0.806 and 0.854.

At LVK's reference point (inspiral truncated at $-40\,t_M$, 220 fit at 10.5 M_f) both analyses find a growth of about +73%, and on LVK's Gaussian statistic they agree, 4.6σ for ours and 4.4σ for theirs. The exact one-sided tail is lower for both, but at this cell it rests on a handful of tail samples, so we do not compare our exact value with theirs. Our A_i is 6.6% larger than LVK's, from more unequal masses (q of 0.713 against 0.879) and lower spins; about a quarter of the gap may come from the mass prior, and the rest is unexplained. The offset bears on the decision. In a post-hoc check prompted by review (status section S11), with our A_i lowered to LVK's value and our ringdown fits unchanged, $P_{\min}$ becomes 0.917 (lane 2: 0.906) and the rule would give the third sentence; removing only the quarter attributable to the prior gives 0.879 (lane 2: 0.865) and the fourth. In both cases lane 1's $P_{\min}$ lies within 2 s.e. (0.022) of the 0.90 boundary. $P_{\min}$ reaches 0.90 when our A_i is 4.0% lower (lane 2: 5.4%), that is, when the excess over LVK falls from 6.6% to about 2.3%, so the choice between the two sentences lies inside the unexplained part of the offset. This post-hoc check does not replace the pre-registered headline. LVK's reference fit is itself 220-only at 10.5 M_f (M_f 72.4 $M_\odot$) and lies above the full-signal remnant; whether it carries the same bias is untested. Prasad (2026) [13], who also splits the signal and models both segments with NRSur7dq4, reports a 4.61σ increase when more than four pre-merger cycles are

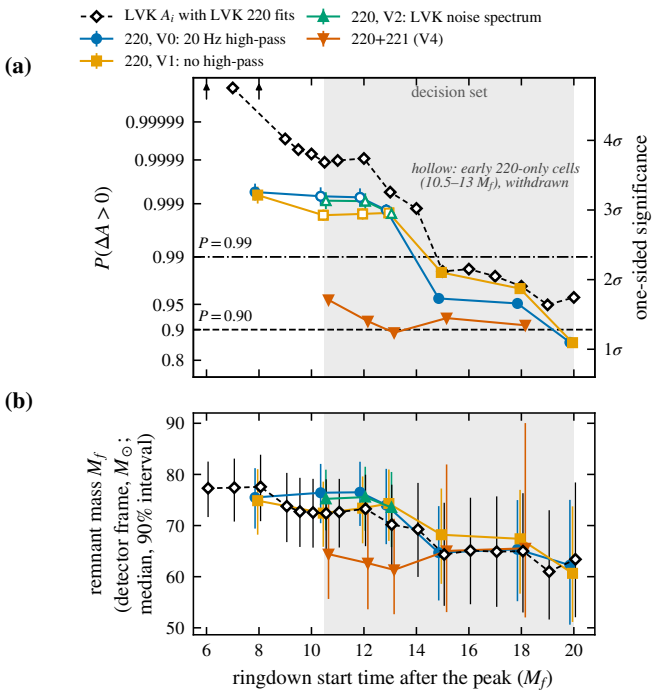


FIG. 4. **GW250114: area evidence and remnant mass against ringdown start time.** (a) $P(\Delta A > 0)$, with one-sided σ on the right axis, against start time for our 220-only variants V0 and V1 (8–20 M_f) and V2 (10.5–13 M_f), our 220+221 fits (V4, 10.5–18 M_f) and LVK's own A_i combined with their 220-only fits; our cells use lane 1's A_i. Lines at $P = 0.99$ and 0.90 mark the pre-registered thresholds and a band marks the decision set. Bars are autocorrelation-aware standard errors of P (i.i.d. at 8 M_f), not systematics, and are smaller than the markers in the decision set; arrows mark the LVK cells at 6 and 8 M_f, where $P = 1$ to sample resolution. The early 220-only cells (10.5–13 M_f, hollow markers) are withdrawn as a single-mode artefact. (b) The remnant mass (median and 90% interval) for the same fits; the 220-only drop between 13 and 15 M_f is absent with the overtone.

excluded. The method reproduces Isi et al. (2021) [6] on GW150914 qualitatively (220 only: $P = 0.946$ against their 95%; with the overtone, 0.911 against 97%, -0.54σ), which constrains it to about $\pm 0.3\sigma$ in P. That calibration was not answer-withheld, because the protocol names the target values and they may be in the agent's training data.

This is not evidence against the area theorem, since every combination favours an increase, nor a high-significance detection, since with the overtone no cell exceeds 0.96 and no retained 220-only cell exceeds 0.982. It does not show that LVK's result is wrong. Several items stay open: the decimation-filter fit for GW250114, which on GW150914 moved M_f by 3.2 $M_\odot$; the overtone fit at 8 M_f, not run; the 220+330 fits, which did not converge (R-hat 1.49–1.72); the cause of the q and spin difference from LVK's posterior; a real rerun at $f_c \pm 10\%$; and the difference between our standard error at 20 M_f (0.0065–0.0090) and the checker's 0.011. On GW150914 the scipy filter lowered M_f, so if it acts the same way here it lowers P and cannot change the sentence; it can only widen the range. Choosing the correct start time or conditioning needs injections, which we have not run. The inspiral and ringdown analyses are disjoint in frequency, not in time, so treating them as independent is an approximation (Methods M3 and Supplementary Section S3).

4. WHAT THE PROTOCOL CAUGHT

The most consequential failure (Table 1) was leakage through the brief. The first Kepler brief prescribed the ellipse and quoted the 8′. No number depended on the leak and the fit was otherwise clean, yet the checker graded it a blocker because it changes what the run can claim. The response was the experiment the protocol had promised, a fit told no model family with the later years withheld, followed by the disguised runs. A smaller leak of the same kind came from the agent itself. The model-agnostic fitter typed three starting magnitudes into its code; the checker found them by reading every numeric literal and showed, by refitting from other starts, that they did not reach the result.

Explanations failed more often than measurements, and in both cases below the measurement stood while the interpretation was withdrawn. The model-agnostic fitter attributed the 2′ offset of the 1600 rows to Tycho's observing series. The data agree with the ephemeris there to half an arcminute, so the offset is not in the data. Only an agent that could see the ephemeris the fitter had been denied could establish this. The checker judged the offset consistent with the limit of a two-body model, an inference it did not test. Separately, the Mercury kill test reported "an error in Newcomb" that Newcomb had corrected in print.

A reading of a measurement failed as well. In episode 5 the checker noted that the strongest cells, with P of 0.998–0.999, sat where a single-mode ringdown fit is known to overestimate the remnant mass; the pre-registered overtone test passed, and those cells were withdrawn. The checker also found that the i.i.d. standard error of P was too small and that our contrast with LVK's exact significance rested on five tail samples. Both were corrected, and the headline sentence did not change.

The remaining failures came from operating conditions. One disguised run never reached the right basin on a machine loaded far beyond its cores. The pre-registered Mercury nuisance model crashed and stalled before the converged fit. Two episode-3 scripts were edited after the results file was written; the checker's re-run showed that the committed code reproduces the committed numbers. On its first day the GW analysis failed to reproduce Isi et al.'s (2019) [14] GW150914 overtone fits, which came out 11–15 $M_\odot$ high; a later session traced most of the gap to the decimation filter and the polarisation model. In episode 5 two cloud submissions failed on script bugs and six simultaneous fits wasted about 5 core-hours to

TABLE 1. **Deviations, failures and corrections.** "Checker" is the re-analysis agent; "arbiter" is the author.

#	ep.	what happened	caught by	consequence
1	1	The prescribed-model brief named the model families and Kepler's 8′ (protocol rule 1).	checker	Run relabelled as a measurement under prescribed models; model-agnostic re-run; deviation recorded.
2	1	The ephemeris verification informed one class-level revision of the quality flags and listed the gross outliers before the fits.	disclosed by the reduction	Outliers kept in the product; disclosed in both fit reports.
3	1	The model-agnostic fitter typed three starting magnitudes.	checker	Disclosed; refits from other starts give the identical solution.
4	1	The 1600 offset was attributed to Tycho's data.	checker	Shown not to be in the data; attributed, by inference, to the model's extrapolation.
5	1	The harmonic baseline used a different index set from the plan.	checker	Disclosed as an implementation change.
6	1	Disguised run beta failed to reach the two-ellipse basin.	the run itself	Reported; declared rerun beta2 succeeded.
7	1	The disguised runs were added after the protocol.	the run record	Labelled post hoc.
8	3	The pre-registered nuisance model is nearly degenerate; the first fit crashed and the main run stalled.	the fitting agent	Converged wide fit gives the headline; post-hoc radial form beside it.
9	3	The headline traced only to the log of a stopped run.	checker	Converged fit writes the headline file.
10	3	Historical fits started from the control's D.	checker	Disclosed; refits from $D = 0$ agree.
11	3	Two model scripts were edited after the results file was written.	checker	Disclosed; the committed code reproduces the committed numbers.
12	3	An "error in Newcomb" was Newcomb's own corrigendum.	checker	Wording corrected; no claim of a new error.
13	5	Day 1: the GW150914 overtone fits did not reproduce Isi et al. 2019 (11–15 $M_\odot$ high).	the analysis agent	Reported; later traced mostly to the decimation filter and polarisation model.
14	5	The pre-registration counted 17 decision cells; its definition gives 15.	the analysis agent	Set used as defined; count corrected.
15	5	The GW150914 calibration of the area method ran after the GW250114 result.	this paper	Recorded here as a deviation; not yet added to PROTOCOL.md.
16	5	Two cloud submits failed; six concurrent fits wasted about 5 core-hours.	the analysis agent	Resubmitted; later jobs ran fewer fits at once.
17	5	The sampler ignored the seed in both inspiral-only runs.	arbiter	Runs identified by checksum; not bit-reproducible.
18	5	The strongest cells (220-only, 10.5–13 M_f) sat where single-mode bias is expected.	checker	Pre-registered overtone test passed; those cells withdrawn; the 10.5–13 M_f range quoted from 220+221 fits.
19	5	The headline "before" posterior was a git-ignored result file.	checker	Posterior samples committed; reruns reproduce every stored value.
20	5	The i.i.d. standard error of P at 20 M_f (0.003) understated the uncertainty (checker 0.011; ours 0.0065–0.0090).	checker	Autocorrelation-aware s.e.; $P = 0.86 \pm 0.01$ at 20 M_f; sentence unchanged.
21	5	The 3.2σ against 3.8σ contrast with LVK rested on five tail samples.	checker	Contrast removed.
22	5	The effect of the unexplained 6.6% A_i offset on the decision rule was not stated.	independent review	Post-hoc rescale (S11): $P_{\min}$ 0.917 (lane 2: 0.906) at LVK's A_i, the third sentence; 0.879 (0.865) at the prior-reweighted A_i, the fourth; stated in Section 3.3; headline unchanged.

contention. The parameter-estimation sampler ignored its seed, so neither inspiral-only run is bit-reproducible. None of these changed a reported number, and each is recorded in Table 1.

The orchestrating agent drafted the briefs, including the one that leaked the shape of the answer. The author's question whether the agent was cheating by recognising Mars led to the disguised test, and the author reported the ignored seed.

5. DISCUSSION AND LIMITATIONS

In episodes 1 and 3 the committed results agree with modern values within their stated uncertainties, except the Mars inclination, whose 3σ excess the checker found consistent with a latitude offset in the reduction. In episode 5 the fractional area growth at LVK's reference point agrees with theirs, and both inspiral-only lanes give the same sentence. Wherever the checker reviewed the work it reproduced the results, by re-running the committed code in episodes 1 and 3 and, in episode 5, by re-deriving the area statistics from the committed posteriors, for the result before the overtone test; a further independent review did the same for the final result (S9

and S10). The ringdown fits were not re-run.

The agents were most useful for work that is slow and error-prone by hand. They transcribed 427 sessions with a confidence on every number and reduced 349 of them, and kept the 187 with good quality flags, at a median residual of 1.38′ against Horizons. In the Mercury analysis, an agent traced a stalled fit to a near-degeneracy between time offsets and Mercury's mean longitude, and in the Tycho reduction the agents flagged three sessions that a one-day shift brings within 2′ (probably date slips). They also declared what they recognised and when. Their weak point was explanation. Twice their interpretations ran ahead of their evidence, and both times a second agent with different information corrected them. In episode 5 a reading of the measurement itself did the same: the strongest cells rested on a known single-mode bias, which the checker flagged and on which the pre-registered overtone test passed. The division of labour that emerged, with agents measuring, a separate agent checking and the author arbitrating, is the method we propose.

The main limit is that the agents know the textbook answers. Withholding the answer, fixing the plan, holding out data and disguising the record narrow what that knowledge can do, and none removes it. The checker is an AI agent started by the author, not an independent expert, and a human domain expert's reading of each measurement is still due. In episode 1 the ephemeris verified the reduction before any fit and produced the list of gross outliers, so the existence of bad rows was known before the fits. The evidence that plans preceded fits rests in episodes 1 and 3 on statements, logs and file times rather than on version history, and in episode 5 the plan preceded the area computation but not the 220-only fits (Section 2). Tycho's transcription and Newcomb's concluded contacts were taken as read; an independent re-transcription and a re-reduction of the about 770 individual Mercury rows remain to be done. Episode 5 leaves the open items listed in Section 3.3 and Supplementary Section S3. The cost was hours of wall time per measurement, about 17 core-hours of rented CPU for the ringdown fits and three multi-hour inspiral-only runs (Methods M6).

6. CONCLUSIONS

Under a written protocol that keeps the expected answer out of the agents' instructions, AI agents re-measured three historical results of physics from the primary records; Section 2 says where the record falls short. Given Tycho Brahe's Mars logs and no model family, an agent chose two Keplerian orbits about one common point, which predicted the withheld years 1596–1601 to a median of 2.46′. Newcomb's Mercury transit contacts of 1677–1881, fitted with a Newtonian model, require an extra perihelion advance of $43.71 \pm 1.04''$ per century, 0.7σ above the 42.98″ general relativity predicts. For GW250114 all 40 cells across both lanes favour growth of the horizon area; once a single-mode bias in the earliest ringdown fits had been found and those cells withdrawn, the retained cells of the headline lane gave $P = 0.86$–0.98, with a minimum of 0.86 at 20 M_f, below the pre-registered 0.90, so the rule withheld the claim. It does not show that LVK's result is wrong; with our A_i lowered to LVK's value, in a post-hoc check, $P_{\min}$ becomes 0.917 and the rule would give the third sentence, "the data favour an increase".

A brief that leaked the shape of the answer, an offset attributed to the wrong cause, the single-mode bias and a failed run were each caught and reported. Wherever the checker reviewed the work it reproduced the results; the ringdown fits were not re-run. The division of labour that emerged, with agents measuring, a separate agent checking and the author arbitrating, is the method we propose.

Its limits are set out in Section 5: the agents know the textbook answers, which the protocol narrows but does not remove, and the checker is an AI agent started by the author, not an independent expert. The next steps are a human domain expert's reading of each measurement, an independent re-transcription of Tycho's logs, a re-reduction of Newcomb's individual Mercury rows and the open items of the GW250114 analysis.

ACKNOWLEDGMENTS

This work was carried out with AI agents built on Claude models (Anthropic), named here by the model the record gives for each role. Claude Opus agents ran the kill tests of episodes 1 to 3, wrote the transcription guide, ran the batch reduction of Tycho's logs and fitted Newcomb's transit contacts. Claude Sonnet 5.5 workers transcribed five of the six chunks of Tycho's logs from the page images, and a Claude Opus agent working in Claude Code transcribed the sixth. A Claude Haiku sub-agent assisted in the kill test of episode 3, and Claude Haiku researchers ran the literature searches. Claude Fable was the second model consulted at review points in episodes 1 and 2 and reviewed drafts of this paper, and Claude Opus agents in Claude Code drafted this text. A Devin agent (Cognition) ran the second inspiral-only analysis of GW250114 on its own machine, independently of the first. The record does not name the models of the fitting and checking agents. The author designed the study, arbitrated every result and is responsible for the content.

DATA AND CODE AVAILABILITY

Code, plans, briefs, reports, checker reports, frozen hashes and run logs are in the paper's repository, github.com/FavioVazquez/tycho-to-gw250114, whose `remeasured/` folder is a byte-identical copy of the time-stamped record: the `road-remeasured` branch of github.com/FavioVazquez/roadtoreality at commit b6b0b05, which contains every file the paper cites, in-

cluding the 131 Hz rule and the episode-5 status sections S9, S10 and S11, except the downloaded public data and 27 of the 54 ringdown posterior sample files (9 MB, among them the 220+221 and 220+330 fits), which are git-ignored. Their sha256 checksums are in the committed `out/CHECKSUMS_outputs.tsv`, the 5%, 50% and 95% quantiles of every fit are in the committed `*_summary.json` files, and the 27 sample files are deposited with their checksums and a snapshot of the branch at commit b6b0b05 in a Zenodo dataset record (10.5281/zenodo.23410453). The episode-5 checker report is kept with the manuscript and answered point by point in status section S10. Data are not committed; each episode has a fetch script and a checksum list. The primary sources are public: Dreyer's *Opera Omnia*, the Principia and the works it draws on, and Newcomb's *Astronomical Papers* in archive.org scans, and the GWOSC strain data and the LVK GW250114 release (Zenodo 10.5281/zenodo.16877102) [15], both under CC BY 4.0. The inspiral-only posterior samples of both GW250114 lanes are committed as CSV files with checksums; the posteriors are not bit-reproducible, because the sampler ignored its seed. Large arrays of the disguised runs and the KeplerBench PDF, public on OpenReview under CC BY 4.0, are not committed. Some scripts contain absolute paths from the machine they ran on.

Appendix A: Methods

This appendix gives the methods of each episode; the main text cites them as M1 to M6.

M1. Tycho's logs: transcription, reduction and fits

Record and transcription. The logs are read from the public scans of J. L. E. Dreyer's *Tychonis Brahe Dani Opera Omnia*, volumes X to XIII [1], which is itself an edited transcription of the manuscripts. Dreyer's footnote corrections were not applied; the printed number is used, and the 18 sessions with such a footnote are flagged. In the kill test an agent transcribed two nights by hand from the page images (the evening of 1595 October 25, six days before opposition, and the morning of 1595 July 22, at quadrature) and reduced them with a script. Against Horizons they agree to 0.87′ and 0.82′, inside the 5′ criterion. Six agent workers then transcribed the Mars sessions in six chunks from a written guide that forbade any ephemeris, any web search for the observations and any computation of positions, and required a confidence on every number and no guessed digits. The guide provides for an independent re-read of about one session in ten and a spot-check by the author against the images; the record does not show whether these were carried out. The comparison of the reduced positions with Horizons (Extended Data Fig. 1) checks the transcriptions as a whole, together with the reduction. The six chunk files hold 87, 68, 69, 67, 66 and 70 sessions, 427 in all, of which 23 hold no Mars data. One session, the evening of 1587 March 27 in the Julian calendar, is traced from the printed page through the crop its worker read and the numbers it recorded to the reduced position and its residual against Horizons (Extended Data Fig. 4).

Reduction. The orchestrating agent reduced each session on its own, from sextant distances, armillary readings, quadrant altitudes and clock times, to geocentric ecliptic longitude and latitude with formal errors, using modern Hipparcos positions for the reference stars, moved to their apparent places at the date, with refraction in the forward model of every reading. Diurnal parallax is applied with a nominal Earth–Mars distance of 1 au, which keeps the ephemeris out of the product at a cost below 0.25′ in the worst case. Of 427 sessions, 23 held no Mars measurement, 55 could not be reduced and 349 were reduced. Internal flags alone sorted the reduced sessions into classes A (37), B (150) and C (162); the flag lists were revised once after a class-level residual table was seen, and never on the basis of a per-session residual. The product is the 187 sessions of classes A and B (sha256 626925e4…). Against Horizons the median total residual is 1.38′; 127 positions lie within 2′ and 177 within 5′; without the five gross outliers above 10′ the RMS is 1.50′ in longitude and 1.78′ in latitude. Tycho's own printed positions, reduced by him, have a median $|\Delta\lambda|$ of 1.89′ against Horizons over 188 entries, a longitude-only statistic rather than the total residual of the product. Class C, excluded on internal flags, has a median of 9.8′.

Prescribed-model fit. The fitter received the 187 positions and the instruction "find the orbit that fits", with Earth's orbit to be fitted from the same data as Kepler did. It was instructed to read only its folder and not to open the repository, any ephemeris or the web, nor to type in any known element as a start or as a check. The brief prescribed the three model families and quoted Kepler's 8′. By the pre-registered BIC rule two Kepler ellipses won ($\Delta\mathrm{BIC} = 9182$). Mars: $a = 1.523697 \pm 0.000065$, $e = 0.093065 \pm 0.000072$, $P = 686.9736 \pm 0.0020$ d. Earth: $e = 0.017003 \pm 0.000091$, $P = 365.2552 \pm 0.0011$ d. A circle with a free equant and the same Keplerian Earth leaves a largest binned median longitude residual of 17.0′ by mean anomaly, against 0.88′ for the ellipses (7.53′ against 1.55′ by apparition). Scaled post hoc to heliocentric longitude near opposition, the equant's error is 3–6′ at the octants, of the order of the 8′ Kepler reported; we did not repeat his construction (a vicarious hypothesis fitted to four acronychal longitudes). The pre-registered 5σ rule rejected 8 rows, among them all five gross outliers. The fit does not model refraction or parallax beyond the reduction. The checker's comparator is a two-body fit to DE441 at the same epochs, the best any two-body model can do. Against it the Mars elements agree within 2σ in a, e, Ω, ϖ, P and mean longitude, and the inclination is 13.7″ (3.0σ) high. From the full covariance the checker obtained the exponent of $P \propto a^p$ as $p = 1.499977 \pm 0.000123$ (formal), $\pm$ 0.000141 after bootstrap scaling.

Model-agnostic fit. The brief described "observed sky positions of one planet" and asked for the model "that predicts it best and is as simple as the data allow". Rows after JD 2304000.5 were withheld. The held-out median is 2.46′ for Kepler ellipses and, with $a^3 \propto P^2$ imposed (post hoc), 2.38′; eccentric circles with equants reach 16.3′, a geocentric deferent with equant and epicycle 22.75′ and the best harmonic series 21′. The geocentric model is handicapped by construction, since its uniform circular epicycle cannot carry the observer's eccentricity, so the conclusion about the equant rests on the heliocentric equant model. With the size ratio free, $a_P/a_E = 1.523649 \pm 0.000062$. The checker reproduced the fit byte for byte and found that no withheld row touched any fit, selection or outlier scale; from the full covariance it obtained $q = 0.666622 \pm 0.000065$ for $a \propto P^q$, the inverse of the exponent p above. The 1600 rows sit about 2′ east of the prediction; those rows agree with DE441 to +0.55′. The fitter had typed starting magnitudes of 0.02 and 0.08 for the eccentricities and 0.03 rad for the inclination; refits from zero eccentricity and random starts reach the identical solution. The withheld set is 28 rows, 15 from the 1600 apparition; one row with a latitude slip dominates the set's 192′ RMS, which should not be quoted. The fitters of the held-out runs were instructed to read only a folder holding their inputs; the checkers could confirm this only from the fitters' statements and the absence of leaked values in the code.

M2. Mercury

The data are Newcomb's 58 concluded contacts (*Astronomical Papers* vol. 1, part VI, 1882, pp. 451–452) [3]: 23 transits, 37 interior contacts and 21 at contact IV, with weights from 0.1 to 6. The agent read them from page images of that table, shown beside the other two records (Extended Data Fig. 5). The kill test found a scatter of about 1–3″ along Mercury's path against Horizons, inside the 10″ criterion. The model is a Newtonian N-body integration of the Sun and planets with DE440/441 masses, started from Horizons states at 1780; the Earth's and the Sun's positions come from Horizons, as the Earth's orbit came from solar observations in Newcomb's time, and Mercury is the system under test. D is carried by an extra inverse-cube central acceleration, which produces a uniform apsidal advance. The fit has ten parameters: Mercury's six elements, D and one constant per contact type. Newcomb's contact times were converted to terrestrial time with the ΔT polynomials of Espenak and Meeus (2006) [16], as the plan fixed; ΔT from Horizons instead is one of the variants in Fig. 3. Extended Data Fig. 3 shows the contact residuals after the headline fit. Robustness variants include Newcomb's own final-solution weighting (43.64 ± 0.89). The leave-one-transit-out jackknife gives a smaller error than the formal one (0.78 against 1.04 in the time form, 0.58 against 0.84 in the radial form); with 23 groups the jackknife's own scatter is about 15%, and we quote the formal, χ^2-scaled value as the conservative one. The 1677 corrigendum is Newcomb's (p. 484: −25.1 s for +29.2 s), and the fit uses his corrected concluded times. Here "answer-withheld" has a narrow meaning: the plan was fixed in advance, the value was never named and no step was tuned to it apart from the start value, disclosed in Section 3.2; the agents' knowledge of the textbook number cannot be excluded. Newcomb's concluded times carry his judgements and rejections; the about 770 individual rows behind them were not re-reduced. Newcomb's 42.95″ rests on a Venus mass of 1/405 000. D absorbs the Sun's oblateness, which is why the control returns 43.01 = 42.98 + 0.03.

M3. GW250114 and the GW150914 calibration

Each Kerr area is

$$A = 8\pi \left(\frac{GM}{c^2}\right)^2 \left(1 + \sqrt{1-\chi^2}\right), \qquad (1)$$

evaluated in the detector frame on both sides so that the redshift cancels in A_f/A_i. A_i comes from two inspiral-only posteriors, both IMRPhenomXPHM at 20–131 Hz with the full likelihood and nested sampling: lane 1 (1000 live points and dlogz 0.1; 4,772 samples) and lane 2, run independently and frozen before it saw lane 1 (by the author's record and the lane-2 merge; 5,046 samples). Both lanes use the same priors: detector-frame chirp mass over 10–100 $M_\odot$ and mass ratio over 0.125–1, both uniform in the component masses (each component constrained to 1–1,000 $M_\odot$), luminosity distance uniform in comoving volume and source-frame time over 50–3,000 Mpc (Planck15 cosmology) and marginalized in the likelihood, spin magnitudes uniform over 0–0.99 with isotropic spin directions, isotropic sky position and orientation, uniform polarization and phase, and a merger time uniform within 0.1 s of the trigger. The 131 Hz cutoff is the (2,2)-mode gravitational-wave frequency at the innermost stable circular orbit of the Kerr remnant whose mass and spin follow from numerical-relativity fits to the median component masses and spins of LVK's full-signal posterior, the rule of LVK's inspiral–merger–ringdown consistency test; recomputed from the LVK posterior file it gives 131.23 Hz for GW250114, and 133.7 Hz against the published 132 Hz on GW150914. A_f comes from our fits with the ringdown package of Isi and Farr [17, 18], 8,000 samples per fit (4 chains × 2,000 draws): 220-only at 8, 10.5, 12, 13, 15, 18 and 20 M_f after the peak, and 220+221 at 10.5, 12, 13, 15 and 18 M_f; 10.5 M_f is the reference start of the LVK release. The sky position, the polarization angle and the reference peak time come from LVK's NRSur7dq4 maximum-likelihood sample (Zenodo 16877102; cited) [15], and the M_f unit for start times uses that sample's final mass, 68.409 $M_\odot$ (detector frame), so a modern remnant mass sets the ringdown timing. The ringdown fits use the strain at the two detectors, shown whitened around the merger beside the other two records

(panel c of Extended Data Fig. 5). Three conditioning variants were fitted for the 220 mode: a 20 Hz high-pass with the autocovariance estimated from the data (V0), no high-pass (V1), and the LVK noise spectrum (V2, at 10.5–13 M_f only); the 220+221 fits (V4) use the V0 settings. Every fit in the decision set uses a remnant-mass prior of 35–150 $M_\odot$ (detector frame), the strain downsampled to 2,048 Hz and 1,000 warm-up draws per chain, and where the autocovariance is estimated from the data (V0, V1, V4) it uses ±128 s of data around the event; the configuration of each fit, including its amplitude prior, is committed in `out/configs/`. $P(\Delta A > 0)$ is exact over all sample pairs (4,772 × 8,000 for lane 1). The inspiral analysis uses the 20–131 Hz band of an 8 s segment and the ringdown fit uses 0.2 s after the peak, above 20 Hz in the high-passed variants, inside the same segment: the two are disjoint in frequency for the signal they model but not in time, the content of the ringdown window below 131 Hz enters both likelihoods, and treating them as independent is an approximation. The pre-registration counted the original decision set as 17 cells; its own definition gives 15, and the set was used as defined.

The final decision set has 20 cells per lane: the 15 220-only cells (V0 and V1 at 10.5–20 M_f, V2 at 10.5–13 M_f) and the five 220+221 cells. For lane 1 the initial area is A_i = 2.99 [2.54, 3.68] $\times 10^{11}\,\mathrm{m}^2$ (median and 90% interval). $P(\Delta A > 0)$ is 0.998–0.999 (2.9–3.2σ one-sided) in the 220-only cells at 10.5–13 M_f, withdrawn as described below; 0.892–0.956 (1.24–1.71σ) in the 220+221 cells at 10.5–13 M_f; 0.911–0.982 at 15 and 18 M_f; and 0.863–0.864 at 20 M_f. Median $\Delta A/A_i$ spans 0.25–0.73 and is positive in every cell. $P_{\min} = 0.863$ at 20 M_f without the high-pass selects the fourth pre-registered sentence; lane 2 gives $P_{\min} = 0.847$ and the same sentence. The pre-registered rule is: $P_{\min} \geq 0.99$ claims the increase (sentences 1 and 2, the second with a decimation-filter qualifier), $0.90 \leq P_{\min} < 0.99$ gives "the data favour an increase", and $P_{\min} < 0.90$ gives the fourth sentence; the sentence templates carry no numbers, which are inserted from the decision set. A 220+330 variant did not converge (R-hat 1.49–1.72, bulk ESS 6–7) and is not in the decision set.

The overtone test was pre-registered with a pass rule: at 10.5–13 M_f the 220+221 median M_f must lie at least 3 $M_\odot$ below the 220-only median, its 13 → 15 M_f drop must be less than half the 220-only drop of 9.3 $M_\odot$, and at 15–18 M_f the two must agree within 2 $M_\odot$. All three held (differences of −12.0, −13.9 and −12.7 $M_\odot$; a 13 → 15 M_f change of +3.7 $M_\odot$; agreement within 0.3 $M_\odot$), so the 13 → 15 M_f drop is accounted for by single-mode bias. With the overtone, M_f is 61–66 $M_\odot$ at every start time, and each 90% interval contains the full-signal detector-frame remnant, 68.06 $M_\odot$.

The ringdown samples are 4 chains of 2,000 draws, so the i.i.d. standard error of P understates the uncertainty, by up to a factor of 2.67. We take the larger of a 100-draw chain-block bootstrap and an effective-sample-size estimate: 0.0065–0.0090 at 20 M_f (effective sample size of the per-sample CDF 786–1,248 of 8,000), and 0.0043–0.0093 for block lengths of 100 to 2,000 draws. The checker's own chain-block estimate is 0.011, and we quote $P = 0.86 \pm 0.01$ at 20 M_f. $P_{\min}$ + 2 s.e. is 0.885 for lane 1 and 0.869 for lane 2, both below 0.90, so by the pre-registered check sentence 4 stands without a qualifier. For the fragility check we counted, in each decision cell, the failing pairs ($A_f \leq A_i$) contributed by the five A_i samples that contribute most. At 10.5 M_f (V0, lane 1) they supply 87% of the failing pairs (A_i of 4.9–5.6 $\times 10^{11}\,\mathrm{m}^2$, against a median of 2.99), and dropping them moves σ from 3.20 to 3.75. In the 220+221 cells they supply at most 2.3% (σ changes by at most 0.011), and in the late 220-only cells at most 5.6% (at most 0.023).

LVK combine an inspiral analysis truncated at $-40\,t_M$ before the peak with the 220 fit at 10.5 M_f and quote the Gaussian significance

$$\frac{\mu_f - \mu_i}{\sqrt{\sigma_f^2 + \sigma_i^2}}. \tag{2}$$

Recomputed from the release files it is 4.447σ, with a minimum of 3.358σ over truncation times, matching the published 4.4σ and 3.4σ. At that reference point both analyses find a growth of about +73%, with 90% intervals of +38% to +109% (ours) and +41% to +116% (LVK), and the Gaussian formula gives 4.6σ for ours and 4.4σ for theirs. We do not compare exact one-sided values at this cell, because ours rests on five tail samples and LVK's samples were not tested for the same fragility. Our A_i is 6.6% larger than LVK's $-40\,t_M$ value (medians of the independent draws used for the swap test, 2.977 against 2.793 $\times 10^{11}\,\mathrm{m}^2$). Both are in the detector frame (median total masses 73.59 and 73.29 $M_\odot$). The difference lies in the posteriors: $q = 0.713$ and spins of 0.21 and 0.27 for lane 1, with lane 2 similar, against $q = 0.879$ and spins of 0.34 and 0.38 for LVK; exchanging masses and spins between the two accounts for the gap. The truncation does not explain it: $-40\,t_M$ corresponds to about 81 Hz on the release median waveform, and no truncation time in LVK's scan reaches our A_i. If LVK's prior is uniform in total mass and mass ratio, reweighting ours to it gives 2.944 $\times 10^{11}\,\mathrm{m}^2$, about a quarter of the gap; the rest, possibly from the waveform and likelihood, is unexplained. Post hoc and prompted by review (status section S11, pre-registered before it was computed), we rescaled the A_i samples of both lanes uniformly, as in the cutoff proxy below, with the ringdown fits unchanged. Lowered by the full offset to LVK's value ($k = 1/1.066$), $P_{\min}$ becomes 0.917 at 20 M_f (lane 2: 0.906), the third sentence; at the prior-reweighted value ($k = 2.944/2.992$) it becomes 0.879 (lane 2: 0.865), the fourth. At lane 1's 20 M_f cells these are 0.9170 and 0.9220 (V0, V1) at LVK's value and 0.8790 and 0.8802 reweighted. $P_{\min}$ reaches 0.90 at $k = 0.960$ for lane 1, an A_i 4.0% lower (lane 2: $k = 0.946$, 5.4% lower), so for lane 1 the excess over LVK would have to fall from 6.6% to about 2.3%. With the 20 M_f s.e. of 0.011, lane 1's $P_{\min}$ is within 2 s.e. of the

0.90 boundary in both cases, as is lane 2's at LVK's value. The pre-registered headline is unchanged. LVK's own A_i with their later 220-only fits gives, by our exact statistic, $P = 0.983$ at 15 M_f and 0.949 at 19 M_f; our rule applied to their scan would give the third sentence ("the data favour an increase"), not the fourth. For the cutoff, a proxy maps 118 and 144 Hz ($f_c \pm 10\%$) to about -16.6 and $-6.9\,t_M$ on LVK's scan, scales our A_i by 1.048 and 1.008 and moves $P_{\min}$ to 0.806 and 0.854, crossing neither 0.90 nor 0.99; neighbouring points of the scan scatter by about ±2%, comparable to these factors. A real rerun would take about 4.3 h of compute for two cutoffs in one lane.

For GW150914, Isi et al. (2021) [6] report $P = 97\%$ with an overtone (220 and 221 at the peak, $N = 1$) and 95% without (220 at +3 ms, $N = 0$). Our "before" posterior is an independent 20–132 Hz IMRPhenomXPHM analysis with 5,142 samples. The pre-registered rule called the method reproducing if $\Phi^{-1}(P)$ agreed within 0.3σ for both cases, and reproducing qualitatively if both agreed within 0.6σ and on the same side of $P = 0.90$. $N = 0$ gives $P = 0.946 \pm 0.002$ (-0.04σ; $\Delta A/A_0$ median 0.63 against 0.60); $N = 1$, with the authors' own 16384 Hz configuration, gives 0.911 ± 0.003 (-0.54σ; 0.42 against 0.52). The verdict is "reproduces qualitatively". The calibration constrains the method to about $\pm 0.3\sigma$ in P and no better: it is one event, our "before" differs from Isi et al.'s time-domain analysis truncated at the peak, and across 19 $N = 1$ settings variants P spans 0.911–0.980, about 0.7σ. A downsampling-filter choice moved the GW150914 remnant mass by 3.2 $M_\odot$ (4.1%). The same test has not been run on GW250114. No combination we computed favours a decrease, so this is not evidence against the area theorem. The closest earlier work is Cabero et al. (2018) [19], Isi et al. (2021) [6], Kastha et al. (2022) [12] and Prasad (2026) [13].

M4. Agents, roles and AI disclosure

The author is the sole author and takes full responsibility for the content. AI systems are not listed as authors. The author set the project's goal, approved the protocol and asked whether the agent was cheating by recognising Mars, the question that led to the disguised test. An orchestrating AI agent drafted the protocol and the briefs, ran the analysis lanes, wrote the transcription guide, merged the transcriptions and ran the reductions. Worker agents transcribed page images and ran the fits. A separate agent, the checker, re-ran each refereed result after its freeze, and a second AI model was consulted at review points. The Acknowledgements name the model behind each role where the record gives one. Agents also ran the literature searches and drafted this text. The author, as arbiter, reviews results and method and does not steer toward the known answer. Who did what in each episode, from the record to the comparison with modern values, is drawn with the counts and times stated in this paper (Extended Data Fig. 6).

M5. Software and data products

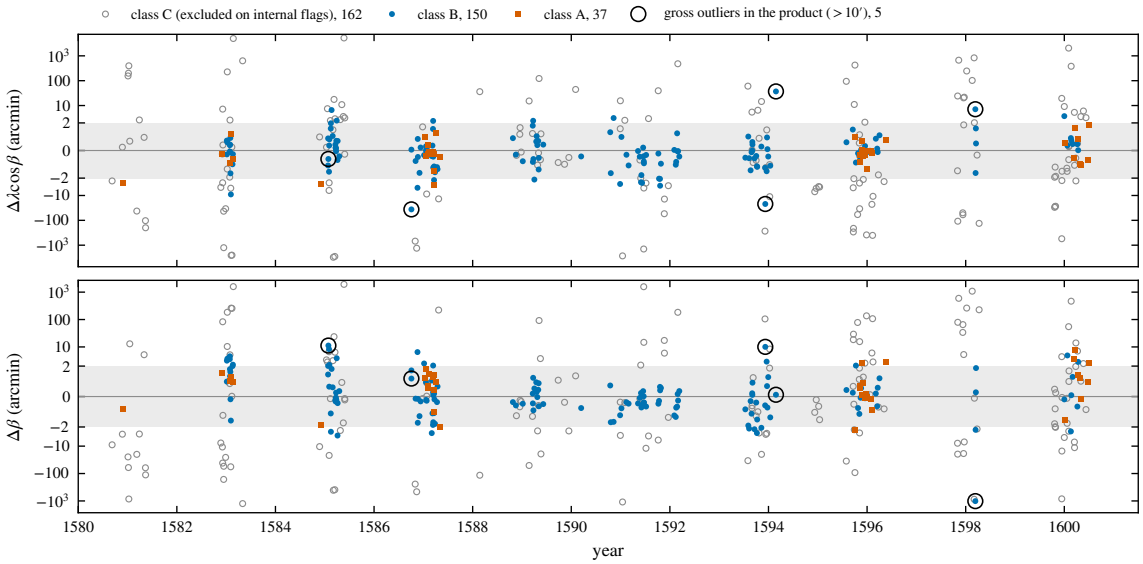
Episode 1 uses numpy [20], scipy [21] and erfa (through Astropy [22]; precession and nutation only), the JPL Horizons service (DE441) for verification and the Hipparcos catalogue for reference stars. Episode 3 uses REBOUND 4.4.10 [23] with the IAS15 integrator [24] and DE440/441 masses. Episode 5 uses the ringdown package 1.1.0 of Isi and Farr [17, 18], bilby 2.8.2 [25] with the dynesty sampler [26], the IMRPhenomXPHM waveform model [27], and the LVK GW250114 data release and GWOSC strain. The figures were made with Matplotlib [28].

M6. Compute and cost

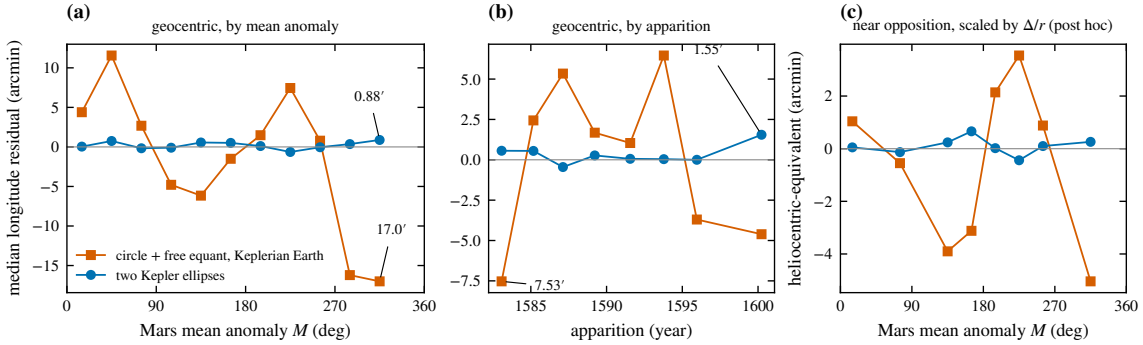
Episodes 1 to 3 ran on one desktop computer, without a cloud instance, at no marginal cost. In episode 1 the kill test took about 25 minutes, the preparation for transcription about 25 minutes and the batch reduction about 55 minutes; the checker's re-runs of the two refereed fits took 5 min 42 s and 16 min 32 s; the transcription and fit sessions have no cost entry yet. Episode 3 took 35 minutes for the kill test, about 65 minutes for the measurement and about 1.5 hours for the fixes the checker required, at $0. In episode 5 the ringdown fits ran on a rented 32-vCPU cloud instance, using about 4.0, 8.1, 0.93 and 2.45 core-hours of CPU in four sessions and 1.4 core-hours in a fifth, for the overtone test; four of these were priced at under $1, $1.5, $1 and $0.5 (on-demand estimates, not bills). The inspiral-only runs took 7770 s on 32 cores (GW250114, lane 1), 6 h 31 min on 8 cores (GW250114, lane 2) and 17933 s on 8 cores (GW150914), and the area combinations ran locally in seconds. Token counts were not available to the agents of episodes 1 and 2, and the episode-5 record gives none; episode 3's cost file records about 190k tokens for one sub-agent in its kill test.

Appendix B: Extended Data figures

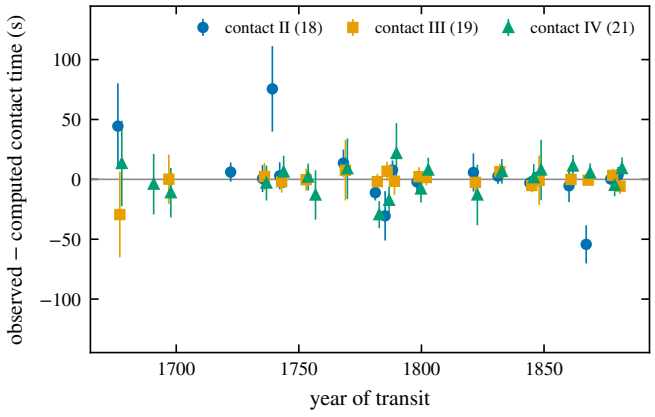
Extended Data Figs. 1 to 6 show the records, residuals and the traced session that support the main text and Appendix A; each is cited where it is used.



Extended Data Fig. 1. **All 349 reduced sessions against the modern ephemeris.** Residuals (reduced minus Horizons, $\Delta\lambda\cos\beta$ and $\Delta\beta$, arcminutes) of every reduced session against date, classes A and B (the 187-position product) filled and class C open, on a symmetric log scale.

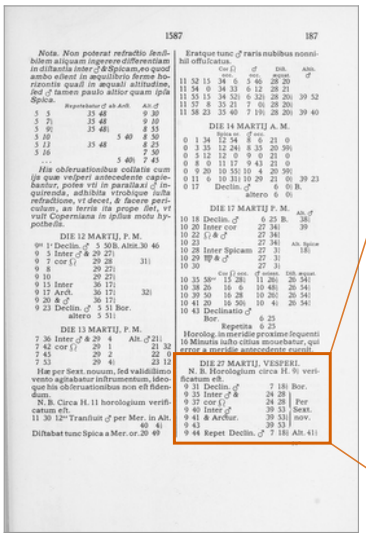


Extended Data Fig. 2. **Kepler's 8′ question (prescribed-model run).** Median longitude residual per 30° bin of Mars's mean anomaly (a) and per apparition (b) for the circle with free equant (same Keplerian Earth) and for two Kepler ellipses; (c) near-opposition residuals scaled to heliocentric equivalents, post hoc. The largest binned medians are 17.0′ against 0.88′ by mean anomaly and 7.53′ against 1.55′ by apparition.



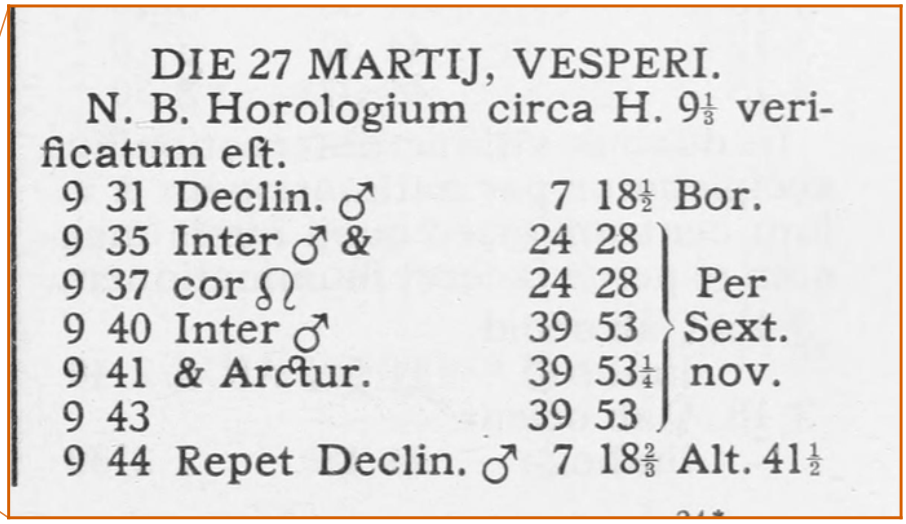
Extended Data Fig. 3. **Mercury: contact residuals after the headline fit.** Observed minus computed time of each of Newcomb's 58 concluded contacts against year, by contact type, for the converged time-offset fit, with 1σ bars $\sigma_0/\sqrt{w}$ from Newcomb's weights w and the fit's $\sigma_0 = 11.29$ s.

a



Dreyer, Opera Omnia vol. XI, p. 187 (scan leaf 201)

b



The crop the worker read, at full resolution and 2× upscaled: x 0.48–0.92, y 0.66–0.83 of the page (data/crops/v11_leaf201_0.48_0.66_0.92_0.83.png)

c

printed line, as the worker recorded it ('|' = column)	number(s) written to the JSON	conf.	reference star: name on page = identification (conf.)
9 31 | Declin. Mars 7 18 1/2 Bor.	declination 7 18½ B	all H	—
9 35 | Inter Mars & cor Leonis | 24 28	distance 24 28 (Sext.)	all H	Cor Leonis = α Leo (Regulus), HIP 49669 (H)
9 37 | 24 28	distance 24 28 (Sext.)	all H	Cor Leonis = α Leo (Regulus), HIP 49669 (H)
9 40 | Inter Mars & Arctur. | 39 53	distance 39 53 (Sext.)	all H	Arcturus = α Boo (Arcturus), HIP 69673 (H)
9 41 | 39 53 1/4	distance 39 53¼ (Sext.)	all H	Arcturus = α Boo (Arcturus), HIP 69673 (H)
9 43 | 39 53	distance 39 53 (Sext.)	all H	Arcturus = α Boo (Arcturus), HIP 69673 (H)
9 44 | Repet. Declin. Mars 7 18 2/3 | Alt. 41 1/2	declination 7 18⅔ B; altitude 41½	all H	—

Session 1587-03-27V in transcriptions/merged.json (heading "DIE 27 MARTIJ, VESPERI."). H: the digits and fraction are unambiguous on the crop. The worker also recorded the clock note and the bracket "Per Sext. nov." (with the new sextant). Distances and declinations are in degrees and arcminutes; B = Borealis (north).

d

Reduced by script, this session alone

7 readings → one position (5 degrees of freedom);

every reading within 0.5′ of the solution

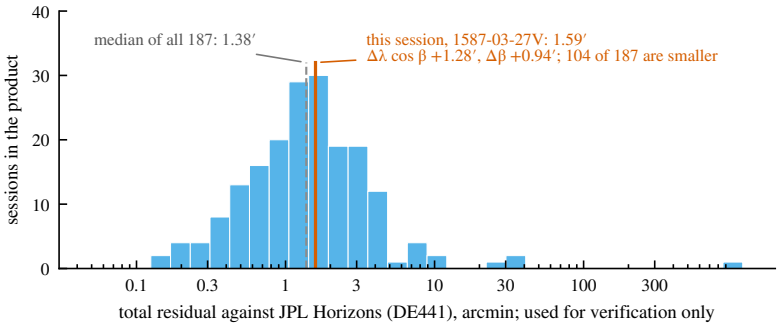
geocentric apparent ecliptic of date, JD(TT) 2300795.3699

longitude $\lambda = 168.4389° \pm 0.54′$

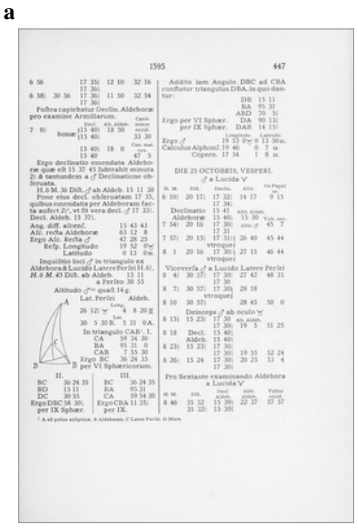
latitude $\beta = +2.9544° \pm 0.71′$

±: formal 1σ (in λ cos β); quality class A

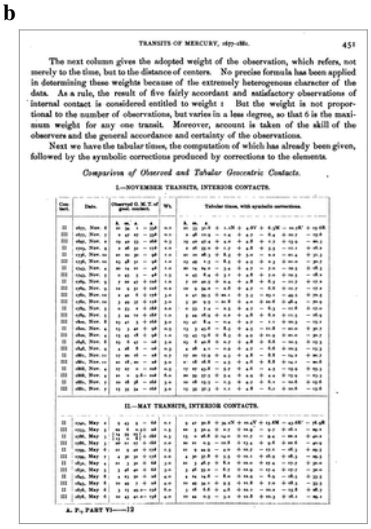
e



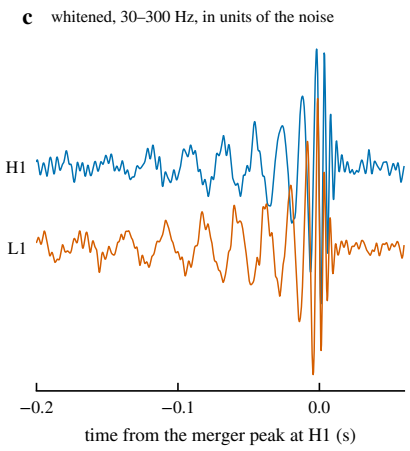
Extended Data Fig. 4. **From page to position: one Tycho session traced end to end.** The evening of 1587 March 27 (Julian calendar), session 1587-03-27V. It was chosen because it is in quality class A, every one of its readings and both star identifications carry the highest confidence, one crop holds the whole session, and its residual lies close to the median, so it is typical rather than the best case. (a) The page as the workers received it, Dreyer's *Tychonis Brahe Dani Opera Omnia* vol. XI, p. 187 (public scan), with the session's block boxed. (b) The crop of that block that the worker read, at full resolution. (c) The worker's transcription: each printed line as recorded, the numbers written to the file with their confidence (H, unambiguous on the crop), and the reference stars as identified, Regulus and Arcturus. (d) The position reduced from this session alone, geocentric apparent ecliptic longitude 168.4389° ± 0.54′ and latitude +2.9544° ± 0.71′ (formal 1σ). (e) Its total residual against Horizons, 1.59′, computed as for Extended Data Fig. 1 (with the true Earth–Mars distance in the parallax), marked on the distribution of all 187 positions of the product (median 1.38′); Horizons was used only for verification.



Episode 1. Tycho's Mars log for 1595, as printed in Dreyer's Opera Omnia (vol. XII, p. 447). Workers received page images and read the numbers on crops.



Episode 3. Newcomb's 1882 comparison of observed and tabular contacts (Astronomical Papers vol. 1, p. 451). The agent read the concluded contacts from page images.



Episode 5. GW250114: public GWOSC strain at the two LIGO detectors. Agents received the strain and the LVK release, and fitted the ringdown to the strain.

Extended Data Fig. 5. **The three records as the agents received them.** (a) Tycho's Mars log for 1595 in Dreyer's *Tychonis Brahe Dani Opera Omnia* vol. XII, p. 447 (public scans), the page of the evening of 1595 October 25; workers received page images and read the numbers on crops. (b) Newcomb's comparison of observed and tabular geocentric contacts, *Astronomical Papers* vol. 1, p. 451 (archive.org scan); an agent read the concluded contacts from page images of pp. 451–452. (c) The GW250114 strain at LIGO Hanford (H1) and Livingston (L1) around the merger, whitened and band-passed to 30–300 Hz, from GWOSC (CC BY 4.0); agents fitted the ringdown to this strain, with the sky position and reference time from the LVK GW250114 release (CC BY 4.0).

	record	workers	reduction / fit	freeze	checker	comparison
Episode 1 Tycho's Mars 1580–1601	Dreyer's edition, vols X–XIII: page images (M1)	6 worker agents, 427 sessions (87, 68, 69, 67, 66, 70); a confidence on every number (M1)	Orchestrating agent reduced 349; 187 kept (classes A, B). Fitting agents: 1 prescribed-model, 1 model-agnostic (28 rows withheld), 4 disguised (3.1, M4)	sha256 of the product and of each fit (FROZEN.txt) (2)	Re-ran the 2 refereed fits: identical (5 min 42 s, 16 min 32 s); no checker for the disguised runs (M6, 2)	Horizons (DE441), verification only: median 1.38′; held-out median 2.46′ (3.1)
Episode 3 Mercury 1677–1881	Newcomb 1882, pp. 451–452: 58 contacts, 23 transits (M2)	One agent: kill test 35 min; contacts scatter about 1–3″ (M2, M6)	Fitting agent: plan 21:17, first fit log 21:35; Newtonian N-body, 10 parameters; about 65 min (2, M2, M6)	Commit order only, no hash (22:13); headline refitted after the checker (2)	Re-ran the code: results file reproduced; converged refit 43.706 ± 1.036; fixes about 1.5 h (3.2, M6)	General relativity, 42.98″ fixed in advance: 43.71 ± 1.04, 0.7σ above (3.2)
Episode 5 GW250114 2025	GWOSC strain and the LVK GW250114 release (M3)	Two independent inspiral-only lanes (4,772 and 5,046 samples); ringdown fits on a rented 32-vCPU instance (M3, M6)	Analysis agent: decision set fixed before any area was computed (S6), after the 220-only fits; 40 cells (2, 3.3)	sha256 of each status version from S6 on; result versions not committed (2)	Report on S1–S8: headline stands, with changes; overtone test, cells withdrawn; a further review recomputed P (2)	LVK, after the result; $P_{\min} = 0.863 < 0.90$: the rule withholds the claim (3.3)

Counts and times as stated in the paper (section in brackets). The author, as arbiter, reviews results and method; the checker is an agent, not an independent expert.

Extended Data Fig. 6. **Who did what, per episode.** One row per measurement, from the record to the comparison with modern values: the record, the workers, the reduction or fit and the agent that ran it, the freeze, the checker and the comparison. Every count and time is one stated in Sections 2 and 3 and Methods sections M1 to M6, and each box names its section. The freeze differs by episode: hashes in episode 1, hashed status versions in episode 5 and commit order alone in episode 3. In episode 5 the decision set was fixed before any area was computed but after the 220-only ringdown fits (Section 2).

Appendix C: Supplementary Information

This appendix holds material that is not needed to follow the main results: a feasibility test (S1), the disguised Kepler runs (S2), open items of the GW250114 analysis (S3) and prior art (S4).

S1. Newton's Moon test and Jupiter's satellites (feasibility test)

This is a kill test only, with no brief file, no pre-registered plan and no checker report, and we do not count it as a measurement. An agent read every number from page images of the Principia (1687 and 1726), Motte's 1729 translation, Picard's *Mesure de la Terre* (1671) and Huygens' *Horologium Oscillatorium* (1673), each with its page reference. With the 1687 inputs (the Moon at 60 Earth radii, a sidereal month of 27 d 7 h 43 m and Picard's 57060 toises per degree) the predicted fall at the Earth's surface in one second is 15.0090 Paris feet, against 15.0957 ft from the seconds pendulum, a ratio of 0.9943 ± 0.025. With the 1726 inputs, which add Newton's solar factor, the ratio is 0.99985 ± 0.025. The uncertainty comes almost entirely from the Moon's distance, taken as ± 0.5 Earth radii. The 1726 agreement is closer than its uncertainty because errors of 0.3 to 1.7% largely cancel. The 1687 printed fall of the Moon in one minute is 0.49% larger than the page's own inputs give. Newton's Moon-test numbers, and whether he adjusted them, are the subject of Westfall (1973) [29] and Kollerstrom (1991) [30]; we have not yet compared our recomputation with theirs. For Jupiter's satellites, a pooled fit of $T \propto a^p$ over the observers Newton tabulates gives $p = 1.498 \pm 0.010$ for the 1726 table (12 degrees of freedom) and the same for 1687; the same method on modern values gives 1.50025 ± 0.00008. The row Newton computed "ex temporibus periodicis" was excluded as a product of the law under test. The 1687 periods are synodic and the 1726 periods sidereal. Two rows may be partly anchored to the law. Cassini's whole-number telescope row understates its rounding error, and the uncertain fraction readings need a human check.

S2. The disguised Kepler runs

After the model-agnostic fit, the frozen product was disguised three ways: time rescaled and shifted, longitude rotated and optionally mirrored, latitude optionally flipped, rows shuffled and every label removed. Fresh agents received "measurements of the direction to one object, as seen by an observer", a withheld tail and a request for a pre-registered plan; no family was named, and the keys stayed private until every run had finished. Alpha, beta2 and gamma recovered the hidden time unit; beta2 and gamma, the runs on the two mirrored copies, also detected the mirror. The exponent p of $P \propto a^p$ came out at 1.50017 ± 0.00014 (beta2) and 1.5001 ± 0.0001 (gamma). Run beta ran on a six-core machine with a load average above 100 and never reached the two-ellipse basin, with in-sample errors of hundreds of arcminutes; we count it as a search failure, not a model judgement. Its rerun, beta2, was declared in advance with the same data and brief and a longer time budget (45 instead of 30 minutes). No checker has reviewed these runs.

TABLE S1. **The disguised-record test.**

run	disguise	outcome	held-out median, unflagged rows
alpha	3.7 d per unit, rotated	two Kepler ellipses	2.05′
beta	0.83 d per unit, rotated, mirrored	failed; no model won	best 1045′ (all rows; none unflagged)
beta2	same data as beta, declared rerun	two Kepler ellipses	2.13′
gamma	12.4 d per unit, rotated, mirrored, latitude flipped	two Kepler ellipses	2.02′

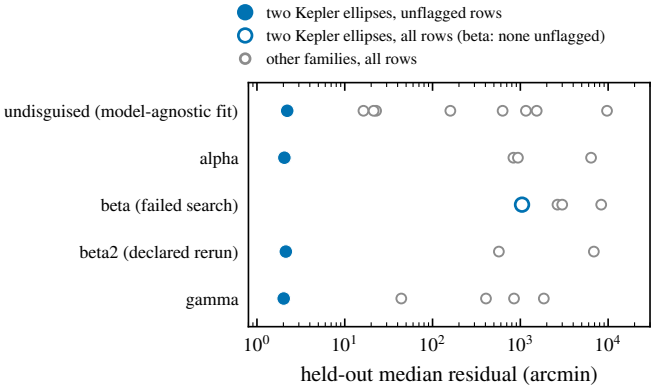


FIG. S1. **The disguised Kepler runs.** Held-out median residual of every family each run tried. The two-ellipse family is drawn at its median over the unflagged withheld rows (alpha 2.05′, beta2 2.13′, gamma 2.02′; the undisguised model-agnostic fit 2.21′), the other families over all withheld rows. Beta has no unflagged row; its best family is drawn at 1045′, the all-row median.

S3. The GW250114 analysis: open items

Every pre-registered step of episode 5 has run and the headline is fixed. The following remain open (status sections S9 and S10): the decimation-filter fit for GW250114, since on GW150914 the downsampling filter moved M_f by 3.2 $M_\odot$; the overtone fit at 8 M_f, outside the decision set, which was not run; the 220+330 fits, which did not converge; the cause of the q and spin difference from LVK's $-40\,t_M$ posterior, which needs new parameter-estimation runs; a real rerun at $f_c \pm 10\%$, about 4.3 h of compute for two cutoffs in one lane; and the difference between our standard error at 20 M_f (0.0065–0.0090) and the

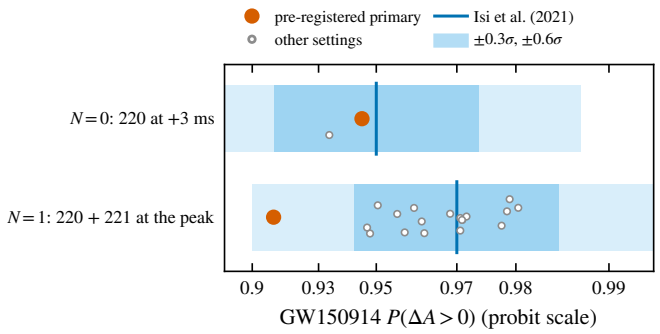


FIG. S2. **The GW150914 calibration of the area method.** $P(\Delta A > 0)$ on a probit scale for the pre-registered primary and the other settings, against Isi et al. (2021)'s [6] 97% ($N = 1$) and 95% ($N = 0$), with the $\pm 0.3\sigma$ and $\pm 0.6\sigma$ bands of the pre-registered rule.

checker's 0.011. On GW150914 the scipy filter lowered M_f, so if it acts the same way here it lowers P and cannot change the sentence; it can only widen the range. Deciding which conditioning or start time is correct needs injections, which have not been run. The checker's report covers status sections S1–S8 and the headline before the overtone test; S9 and S10 were run in answer to it. A further independent review, by another re-analysis agent, then recomputed from the committed samples P for the 20 M_f and 220+221 cells, the lane A_i statistics and the f_c proxy, plus the fragility and standard-error figures. The ringdown fits were not re-run. The full scan tables are in `ep5-gw250114-area/out/STATUS.md`, sections S6, S7, S9, S10 and S11 (post hoc), and in `results/area_task4.csv`, `results/area_task8_checks.csv`, `results/area_task8_blocklen.csv` and `results/area_task9_scale.csv`.

S4. Prior art

For the Kepler episode, Li, Ji and Zhang (arXiv:2111.12210) [31] recover Kepler's laws from 28 heliocentric Mars positions that Kepler had already computed, taken from *Astronomia nova* rather than Tycho's logs. KeplerBench (Malhotra 2026) [4] is a benchmark for discovering physical laws from observer-centred astronomical measurements without privileged variables, scored on historical (Tycho) and synthetic star-system evidence at three levels: prediction, global law and local law. Its pilot uses Tycho's Mars declinations as transcribed by Pafko, reduced to 354 nightly epochs from 1582 to 1600, with the 268 epochs before 1594 for training and the 86 of 1594–1600 held out. A six-parameter ellipse that assumes a heliocentric orbit and the modern Earth orbit degrades from 2.66′ in sample to 13.31′ held out, while a flexible curve fit misses the later campaigns by three orders of magnitude. The paper notes that the pilot does not itself establish discovery, since the ellipse was supplied. Our model-agnostic result is qualitatively the same; what it adds is stated in Section 1. Pafko's transcription [5] of the Mars entries serves visualisation, without a fit. Krisciunas (arXiv:1610.05750) [32] repeated the measurement with his own naked-eye observations. Carman (2021; 2025a, b, c) [33–36] and Carman and Recio (2022) [37] reconstruct how Tycho, Longomontanus and Kepler computed with the observations. Wesley (1978) [8] assesses the accuracy of Tycho's instruments. We could not find a night-by-night reduction of the raw readings checked against a modern ephemeris.

For Newton, Westfall (1973) [29] and Kollerstrom (1991) [30] discuss Newton's Moon-test numbers, and Aoki (1992, 1996) [38, 39] studies the accuracy of the test as printed; we could read only the records of Aoki's papers. Longhin (2025) [40] fits Galileo's satellite observations, not Newton's table. We could not find a published fit, with uncertainties, of the period–distance exponent to the Principia's own Jupiter table.

For Mercury, Morrison and Ward (1975) [10] fitted about 2400 internal-contact times of 1677–1973 with the analytical theories of their day and found $+41.9 \pm 0.5''$ per century. Constantin (2011; arXiv:1104.0548) [41] refitted Le Verrier's observations, obtaining 42.8″ per century against Le Verrier's 38.3″. Duncombe (1956) [42] quotes an observed-minus-theory excess of $43.11 \pm 0.45''$ per century (43.03 from relativity). For the area law, Cabero et al. (2018) [19] defined the separate inspiral and ringdown area test, Isi et al. (2021) [6] applied it to GW150914, Kastha et al. (2022) [12] showed that single-mode ringdown fits overestimate the final mass and inflate the confidence, and Prasad (2026) [13] and Chung et al. (2026) [43] are independent area analyses of GW250114.

On machines and the history of science, AI Feynman (2020) [44] and AI-Descartes (2023 [45], which recovers Kepler's third law) study the rediscovery of known laws by machine, mostly from prepared or synthetic data. We could not find agents run on raw historical measurement records.

REFERENCES

[1] J. L. E. Dreyer, ed., *Tychonis Brahe Dani Opera Omnia* (Libraria Gyldendaliana, Copenhagen, 1913–1929) vols. X–XIII (observations).
[2] U. J. Le Verrier, Lettre de M. Le Verrier à M. Faye sur la théorie de Mercure et sur le mouvement du périhélie de cette planète, Comptes Rendus Hebdomadaires des Séances de l'Académie des Sciences **49**, 379 (1859).
[3] S. Newcomb, Discussion and results of observations on transits of Mercury from 1677 to 1881, in *Astronomical Papers Prepared for the Use of the American Ephemeris and Nautical Almanac*, Vol. 1 (U.S. Nautical Almanac Office, Washington, 1882) part VI.
[4] A. Malhotra, KeplerBench: A Benchmark for Scientific Discovery from Observations, NeurIPS 2026 Workshop Sim2Sci, OpenReview RdErwlc7jJ (2026), CC BY 4.0.
[5] W. Pafko, Tycho Brahe's Mars observations, transcription of the Mars entries of the *Opera Omnia* (2000).
[6] M. Isi, W. M. Farr, M. Giesler, M. A. Scheel, and S. A. Teukolsky, Testing the Black-Hole Area Law with

GW150914, Physical Review Letters **127**, 011103 (2021), arXiv:2012.04486.
[7] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes, Physical Review Letters **135**, 111403 (2025), arXiv:2509.08054.
[8] W. G. Wesley, The Accuracy of Tycho Brahe's Instruments, Journal for the History of Astronomy **9**, 42 (1978).
[9] C. M. Will, The Confrontation between General Relativity and Experiment, Living Reviews in Relativity **17**, 4 (2014).
[10] L. V. Morrison and C. G. Ward, An Analysis of the Transits of Mercury: 1677–1973, Monthly Notices of the Royal Astronomical Society **173**, 183 (1975).
[11] S. W. Hawking, Gravitational radiation from colliding black holes, Physical Review Letters **26**, 1344 (1971).
[12] S. Kastha, C. D. Capano, J. Westerweck, M. Cabero, B. Krishnan, and A. B. Nielsen, Model systematics in time domain tests of binary black hole evolution, Physical Review D **105**, 064042 (2022), arXiv:2111.13664.
[13] V. Prasad, Multi-segment consistency tests of general relativity (2026), arXiv:2603.05835.
[14] M. Isi, M. Giesler, W. M. Farr, M. A. Scheel, and S. A. Teukolsky, Testing the No-Hair Theorem with GW150914, Physical Review Letters **123**, 111102 (2019), arXiv:1905.00869.
[15] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, GW250114 discovery paper figure scripts, Zenodo data release (2025), CC BY 4.0.
[16] F. Espenak and J. Meeus, *Five Millennium Canon of Solar Eclipses: −1999 to +3000 (2000 BCE to 3000 CE)*, NASA Technical Publication NASA/TP-2006-214141 (NASA Goddard Space Flight Center, 2006).
[17] M. Isi and W. M. Farr, Analyzing black-hole ringdowns (2021), arXiv:2107.05609.
[18] M. Isi, W. M. Farr, *et al.*, maxisi/ringdown, Zenodo software record, concept DOI for all versions (v1.1.0 used here) (2024).
[19] M. Cabero, C. D. Capano, O. Fischer-Birnholtz, B. Krishnan, A. B. Nielsen, A. H. Nitz, and C. M. Biwer, Observational tests of the black hole area increase law, Physical Review D **97**, 124069 (2018), arXiv:1711.09073.
[20] C. R. Harris, K. J. Millman, S. J. van der Walt, *et al.*, Array programming with NumPy, Nature **585**, 357 (2020).
[21] P. Virtanen, R. Gommers, T. E. Oliphant, *et al.*, SciPy 1.0: fundamental algorithms for scientific computing in Python, Nature Methods **17**, 261 (2020).
[22] Astropy Collaboration, The Astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package, The Astrophysical Journal **935**, 167 (2022).
[23] H. Rein and S.-F. Liu, REBOUND: an open-source multi-purpose N-body code for collisional dynamics, Astronomy & Astrophysics **537**, A128 (2012).
[24] H. Rein and D. S. Spiegel, IAS15: a fast, adaptive, high-order integrator for gravitational dynamics, accurate to machine precision over a billion orbits, Monthly Notices of the Royal Astronomical Society **446**, 1424 (2015).
[25] G. Ashton *et al.*, BILBY: a user-friendly Bayesian inference library for gravitational-wave astronomy, The Astrophysical Journal Supplement Series **241**, 27 (2019).
[26] J. S. Speagle, DYNESTY: a dynamic nested sampling package for estimating Bayesian posteriors and evidences, Monthly Notices of the Royal Astronomical Society **493**, 3132 (2020).
[27] G. Pratten *et al.*, Computationally efficient models for the dominant and subdominant harmonic modes of precessing binary black holes, Physical Review D **103**, 104056 (2021).
[28] J. D. Hunter, Matplotlib: A 2D graphics environment, Computing in Science & Engineering **9**, 90 (2007).
[29] R. S. Westfall, Newton and the Fudge Factor, Science **179**, 751 (1973).
[30] N. Kollerstrom, Newton's two 'moon-tests', British Journal for the History of Science **24**, 369 (1991).
[31] Z. Li, J. Ji, and Y. Zhang, From Kepler to Newton: Explainable AI for Science (2021), arXiv:2111.12210.
[32] K. Krisciunas, Demonstrating the Elliptical Orbit of Mars using Naked Eye Data (2016), arXiv:1610.05750.
[33] C. C. Carman, The Tychonic Method for Calculating the Ratio between the Eccentricities of Mars, Journal for the History of Astronomy **52**, 212 (2021).
[34] C. C. Carman, When genius met data: Kepler's first exploration of Tycho's observations, Archive for History of Exact Sciences **79**, 6 (2025).
[35] C. C. Carman, Did Tycho (Or His Assistants) manipulate his observations? An analysis of Tycho's table of mars mean oppositions, Historia Mathematica **73**, 103187 (2025).
[36] C. C. Carman, Mars's models behind Kepler's ephemerides of 1604 and 1605, Archive for History of Exact Sciences **79**, 16 (2025).
[37] C. C. Carman and G. L. Recio, Tycho Brahe's Appendix ad Observationes anni 1593 and the date of Brahe's theory of Mars, the prototype for Kepler's vicarious hypothesis, Journal for the History of Astronomy **53**, 239 (2022).
[38] S. Aoki, The moon-test in Newton's Principia: Accuracy of inverse-square law of universal gravitation, Archive for History of Exact Sciences **44**, 147 (1992).
[39] S. Aoki, Corrections and additions for "the moon-test in Newton's Principia: Accuracy of inverse-square law of universal gravitation", Archive for History of Exact Sciences **49**, 393 (1996).
[40] A. Longhin, A quantitative analysis of Galilei's observations of Jupiter satellites from the Sidereus Nuncius (2025), arXiv:2503.12543.
[41] D. R. Constantin, On the first determination of Mercury's perihelion advance (2011), arXiv:1104.0548.
[42] R. L. Duncombe, Relativity effects for the three inner planets, The Astronomical Journal **61**, 174 (1956).
[43] A. K.-W. Chung, K. K.-H. Lam, A. Liu, and N. Yunes, Measuring a Black Hole's Area Immediately after Merger: A Direct-Wave Test of Hawking's Area Law (2026), arXiv:2606.06592.
[44] S.-M. Udrescu and M. Tegmark, AI Feynman: A physics-inspired method for symbolic regression, Science Advances **6**, eaay2631 (2020).
[45] C. Cornelio, S. Dash, V. Austel, T. R. Josephson, J. Goncalves, K. L. Clarkson, N. Megiddo, B. El Khadir, and L. Horesh, Combining data and theory for derivable scientific discovery with AI-Descartes, Nature Communications **14**, 1777 (2023).